She Empower



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Bachelor of Science in IT (2022-2026)

Department of Computer Science & Information Technology

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She Empower: Voice Guided Marketplace App

A project presented to

University of the Punjab, Lahore

In partial fulfillment of the
requirement for the degree of

Bachelor of Science in IT (2022-2026)

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CERTIFICATE OF APPROVAL

It is to certify that Iman Ishaq [076991] and Ramsa Afzal [077021] developed the final year design project (FYDP) of BSIT "She Empower" under the supervision of "**Prof. Usman Ahmed**". In my opinion, it is fully adequate, in scope and quality for the degree of Bachelors of Science in IT.

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Executive Summary

Women often struggle to use their skills and start their business due to limited market access, Guidance, and complex interfaces. To solve this issue, 'She Empower', an Android application, was built using Java in Android Studio with a Firebase database. This app will enable women to create personalized profiles and display their skills for sale. The platform will allow sellers to promote their Different talents, including cooking, stitching, art, handmade crafts, and jewelry making, etc. While buyers can discover and purchase these skill-based services. With four simple panels (Seller, Buyer, Admin, and Visitor) and voice-guided Urdu support, the app will help women to launch and manage their skill-based businesses independently without technical expertise. Buyers can browse services, place orders using a unified order form, and upload payment proofs via Easypaisa or JazzCash. Orders are auto-accepted, and sellers can update the order status through "Started → In Progress → Completed for Delivery/Pickup" stages, with the buyer receiving real-time notifications. Testing confirmed that the app loads in under 3 seconds. Buyers and sellers can contact each other directly through WhatsApp for communication regarding orders and services. She Empower successfully empowers local women by giving them a safe, simple, and accessible digital marketplace.

Keywords: Voice-guided marketplace, women entrepreneurship, Android,

Firebase, Urdu TTS, manual payment verification.

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Abbreviations

All the abbreviations and acronyms used in our FYP along with their respective brief description.

Abbreviations	Full Form	Description
APK	Android Package Kit	File format used to distribute and install Android applications.
API	Application Programming Interface	A set of protocols that allow different software systems to communicate.
CNIC	Computerized National Identity Card	The official identity document of Pakistani citizens.
FCM	Firebase Cloud Messaging	Google's service for sending push notifications to Android devices.
FR	Functional Requirement	A specific behavior or function the system must perform.
FT	Functional Testing	Testing to verify that the system functions according to requirements.
IT	Integration Testing	Testing to verify that different modules of the system work together correctly.
JSON	JavaScript Object Notation	A lightweight data-interchange format.
SDK	Software Development Kit	A set of tools for building software applications.
TTS	Text-to-Speech	Technology that converts text into spoken voice output (used for Urdu voice guidance).
UC	Use Case	How a user interacts with a system to achieve a goal.
UI	User Interface	A system that users interact with visually.
UT	Unit Test	Testing a single function or component of code.
UML	Unified modified Language	Way to visualize and design software-using diagrams.

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Chapter 1: Introduction



1. Introduction

This chapter provides overview of the "She Empower" project. It explains the project's background. It describes the main problem that local areas women face and how this project will help to solve it. It further outlines the proposed solution, the project's goals, and defines the boundaries of its scope. Additionally, this chapter presents an analysis of related systems, the project's vision statement, constraints, and execution method.

1.1. Problem Statement

Many women in rural areas have good skills like cooking, stitching, and making handicrafts. These women want to earn money but cannot find an easy way to sell their services online. The problem is not their skills. The problem is that existing apps are too hard to use. Apps like Daraz, Fiverr, and Etsy have difficult screens and long forms. These apps made for people who already know how to use technology well. Most local women do not have this technical knowledge or training. The apps also do not provide help in the Urdu language. Because of these problems, almost half of the skilled women cannot start their own online business. There is a clear need for an app that is simple and speaks to the user in Urdu. Right now, no easy, voice-guided platform exists to help these women sell their skills.

1.2. Problem Solution

This project will create an Android app "She Empower" to solve this problem. The app will use Urdu voice guidance to tell users exactly what to do at each step. Women will not need to read long instructions. The app will speak to them. The app will have four simple sections for Sellers, Buyers, Admin, and Visitors. Sellers can easily create a profile and show their skills. Buyers can easily find services and place orders. The app will not use difficult online payment systems. Instead, buyers will upload a simple screenshot of their payment. Admin will check this screenshot to confirm the payment.

For safety, only women can register as sellers by showing their CNIC and a voice sample.

1.3. Objectives of the Proposed System

- Develop a voice-guided app that provides Urdu voice instructions on every screen.
- Create four separate panels for Sellers, Buyers, Admin, and Visitors so each user sees only what they need.
- Set up a system where women sellers must upload a CNIC picture and a voice sample for admin approval before they can sell.
- Provide a simple WhatsApp contact option so buyers and sellers can communicate directly through WhatsApp.
- Test app with real users and target 70% positive feedback that how helpful is it.

1.4. Scope

The scope of this project is to develop an Android App specifically for local women. App will develop by using Java to ensure a simple interface, and Firebase will used for database flexibility. The project will not include live payment gateway integration. The first release will focus on user registration, profile management for sellers and buyers, product listing, order management system, and payments by proof. Development will be divided into: Project proposal, RTM (Requirement Traceability Matrix), SRS (Software Design Specification), Design (UI/UX design) , Implementation, Testing, Evaluation and Deployment.

Out-of-Scope/Constraints:

- **No Automated Payment Gateway Integration:** The project will not include integration with live credit/debit card. All payment verification will be manual.
- **Platform Limitation:** The app will be developed only for the Android OS.

1.5. System Components:

The "She Empower" app is structured around these primary modules:

1.5.1. Module-1: Frontend UI (Android App)

- Built using Java in Android Studio *and* renders all user interface screens.
- Implement Google Text-to-Speech (TTS) for Urdu voice-guidance on every screen.
- Handles all user interactions (clicks, form submissions, navigation).
- Communicates with backend services through REST calls and Firebase SDKs.

1.5.2. Module-2: Authentication Service

- Manages secure user registration (email/password).
- Handles login verification, session management and password reset functionality.
- Built on Firebase Authentication for secure identity verification.
- Assigns user roles after registration.
- Includes email verification and password strength validation.
- Features session timeout mechanisms for security.
- Ensures users only access features relevant to their assigned role.
- Only sellers can access skill management interfaces.
- Only admins can access the verification dashboard.

1.5.3. Module-3: User Service

- Creates and updates buyer and seller profiles.
- Stores personal information (name, phone, email, preferences).
- Handles storage of seller verification documents (CNIC images, voice samples).
- Interacts with Firebase Firestore for data persistence.
- Uses Firebase Storage for media file management.
- Manages profile picture uploads and management.
- Users can only modify their own profile data.

1.5.4. Module-4: Order Service

- Manages complete order lifecycle (creation, acceptance, completion).
- Handles order creation and placement by buyers.
- Stores orders in Firestore database and maintains comprehensive order record.
- Enables status changes: Accepted → In Progress → Completed.
- Sends instant notifications to buyers for each status change.
- Enables review of past transactions.

1.5.5. Module-5: Seller Service

- Enables women to manage their online business.
- Allows creation and management of service listings and availability management.
- Manages service titles, descriptions, pricing, and sample images.
- Provides seller dashboard for real-time order monitoring.
- Enables sellers to view all received orders and updating service descriptions.
- Tracks seller performance through ratings and reviews.
- Monitors earnings from completed orders.
- Provides a WhatsApp contact option for direct communication with buyers.
- Facilitates discussions about order customization and delivery through WhatsApp.

1.5.6. Module-6: Payment Service

- Manages proof-based manual payment verification system.
- Eliminates need for complex online payment gateway integration.
- Accepts payment proofs from mobile wallets.
- Buyers upload payment proof screenshots.
- Stores payment proof images in Firebase Storage.
- Tracks payment verification status (Pending, Approved, and Rejected).
- Maintains complete transaction history.
- Implements Firebase Security Rules for data protection.

1.5.7. Module-7: Admin Service

- Provides comprehensive administrative controls.
- Admins review and approve seller registrations by examining.
- Ensures only verified women can operate on platform.
- Enables payment proof verification.
- Allows admins to review uploaded payment screenshots.
- Confirms transaction details match order information.
- Enables platform settings management.
- Monitors system performance and handles platform disputes.
- Maintains platform security by tracking suspicious activity.

1.5.8. Module-8: WhatsApp Contact Service

- Provides direct communication between buyers and sellers through WhatsApp.
- Opens WhatsApp from the application using seller's registered phone number.
- Allows buyers to contact sellers regarding services and orders.
- No messages are stored inside the She Empower application.
- No in-app chat history is maintained.
- Firebase Firestore is not used for storing chat messages.
- WhatsApp handles message delivery and conversation history.

1.5.9. Module-9: Firebase Database (Firestore & Storage)

- Serves as central data persistence layer.
- Utilizes Google Firebase's comprehensive services.
- Provides real-time data synchronization.
- All clients receive instant updates when data changes.
- All services interact through standardized SDKs and API.
- Features automatic backup and ensures global content availability.
- Implements offline persistence feature by Firebase Security Rules.

1.6. Related System Analysis/Literature Review:

Analysis of the existing platform shows a clear gap that "She Empower" is designed to fill.

Table 1 Related System Analysis and Literature Review

Application Name	Weakness	Proposed Project Solution
Daraz/Amazon	A difficult seller registration interface. Users require technical training to navigate the platform. No voiceguidance is available.	Provides a simplified registration interface with minimal steps and systematic Urdu voice-guidance.
Etsy	Globally focused on international shipping with currency exchange issues.	Designed and supports local manual payment methods (Cash on Delivery,JazzCash) with proof-based verification.
Fiverr/Upwork	Freelancing marketplace better for digital skills for selling and buying. Accounts can be suspended for minor policy infractions, creating instability for sellers.	Specifically designed for traditional skills like, cooking, stitching, art, handmade crafts. The auto- acceptance model ensures that sellers never face account suspensions.
Facebook groups/Marketplace	Lack structured order management and payment verification systems. No dedicated seller verification process, raising privacy concerns for women. Communication is unorganized and no formal transaction record.	Provides structured order management, direct WhatsApp communication between buyers and sellers. Offers secure women only seller community verified through CNIC and voice sample verification.

1.7. Vision Statement

For [Talented women in local and rural areas] **Who** [needs a simple, voice-guided app to sell their services online?] **The** [She Empower] **Is** [an Android marketplace] **That** [provides Urdu voice-guidance, easy order management, and a simple interface] **Unlike** [complex platforms] **Our platform** [offers women-focused design and a manual verification system, enabling them to start and run an online business with confidence.]

1.8. System Limitations and Constraints

The project works under the following limitations and constraints:

Limitations (External Factors):

- Voice guidance and app performance may vary on devices with an Android version below API Level 21 (Android 5.0 Lollipop).
- Initially, the app will designed for local areas and not tested for other cities.
- Payment Verifications are manuals. This can cause delays in order confirmation.
- Without the Internet, users cannot place orders and send or receive messages.

Constraints:

- **Performance:** App must load within 3 seconds.
- **Security:** User data (CNIC, phone, email) must be encrypted using Firebase rules.
- **Usability:** Urdu voice-guidance provide on every screen for users.
- **Timeline:** Project completed within 5-6 months with a team size of 2 members.
- **Budget:** Only free tools will use (Android Studio, Firebase free tier, Google TTS).

1.9. Tools and Technologies

The development of "She Empower" will use the following tools and technologies:

Table 2 Tools and Technologies for Proposed Project

	Tools	Version	Rationale
Tools, Libraries, and Technologies	Android studio	2025.2.1	Official IDE for Android Development with emulators and debugging tools.
	Git and GitHub	2.51.2	Version control for tracking code changes and collaboration between team members.
	Technologies	**Version**	**Rationale**
	Firebase SDK	v34.12.0.	Backend as a Service for real-time database authentication and cloud storage for images and voice samples.
	Google TTS	Latest	Library used to implement core feature of Urdu voice guidance and speech output for navigation.
	Java	SE 25.0.1/ 2.3.20	Main programming languages for Android application logic and UI functionality.

1.10. Project Deliverables

The following key documents will be delivered as part of this Final Year Design Project:

1. Project Planning Document

Detailed plan including scope, objectives, timeline, milestones, and task allocation between two team members (Ramsa Afzal and Iman Ishaq).

2. Software Requirements Specification (SRS)

Comprehensive document covering functional and non-functional requirements, user roles (Seller, Buyer, Admin, Visitor), use case descriptions, and traceability matrix.

3. System Design Document (SDS)

Architectural design (MVC pattern), component diagram, class diagram, state transition diagram, sequence diagrams, and data dictionary for Firebase Firestore collections.

4. Working Prototype

A functional Android application-demonstrating core features. User registration and login (role-based via Firebase Authentication)

Seller: Register with CNIC and voice sample, add skills, view orders, update order status,and receive whatapp contact through whatsapp.

Buyer: Browse categories, search services, place orders, upload payment proof and contact sellers through WhatsApp.

Admin: Verify seller registrations, verify payment proofs, and manage users

Visitor: Browse services and categories without account.

5. Source Code Repository

Complete Android (Java) source code with Firebase configuration, hosted on GitHub, with a README file containing setup and installation instructions.

6. Testing and Evaluation Document

Test cases (unit, functional, integration, performance), test results, performance benchmarks (app load time, chat latency, concurrent order handling), and security testing (Firebase rules, data access restrictions).

7. User Manual

Step by step guide for sellers and buyers: how to register, upload CNIC and voice sample, add skills, place orders, upload payment proof, contact sellers through WhatsApp and track order status — with Urdu voice guidance instructions.

8. Deployment Guide

A document outlining how to build the signed APK in Android Studio, configure Firebase production environment, set security rules, deploy using Docker emulator, and distribute the APK to end users.

9. Final Project Report

Complete academic document summarizing introduction, literature review, system analysis, design, implementation, testing results, and future work.

10. Appendices

Additional materials: use case descriptions (fully dressed format), coding standards, application prototype screenshots (Figma/Canva UI designs), and references.

1.11. Project Planning

A Gantt chart being the plans for carrying out the project activities, representing the milestones as well as resources needed for each task:

Table 3 Project Planning

ID	Task Name	Duration						
		Week 1-3	Week 4-7	Week 7-112	Week 12-15	Week 15-19	Week 19- 22	Week 22-25
1	Requirement Analysis							
2	Frontend							
3	Backend							
4	Database							
5	APIs							
6	Testing							
7	Final Deployment							

1.12. Summary

This first chapter sets the foundation for the "She Empower" project. It shows a severe problem faced by skilled women in local areas. It also presents a solution, which is a voiceguided mobile app made just for women. The project has clear goals that focus on making the app easy to access, secure, and satisfying to use. The project scope is to ensure a highquality result. It also helps finish the work on time for the academic schedule. By looking at existing platforms, the project found a major gap in the market. This gap has been checked and confirmed. The project aims to fill that gap. The next chapters will explain the careful analysis, design, building, and testing steps.

Chapter 2: Analysis



2. Analysis

This chapter presents analysis of requirements for the She Empower app. It shows the user classes and their characteristics, describes the techniques used to gather requirements, and explains both functional and non-functional requirements. External interface requirements are document to ensure communication between system and its users.

2.1 User Classes and Characteristics:

This table shows the types of users who will use this app and explains their key features:

Table 4 User Classes and Characteristics

User Classes	Class Characteristics
Seller	Women with skills like cooking, stitching, handmade crafts, jewelry have low digital knowledge so this app uses Urdu voice-guidance. They must register with CNIC image and voice samples. Can add/edit/remove skills, update order-status and chat with buyers.
Buyer	Customers who want real local products. Have basic smartphone knowledge. Can browse categories, search services, place orders with single form, select delivery and payment method, and see order history.
Admin	Runs the system and knows how it works. Check seller's verification (CNIC + voice), payment proofs, and platform activity. Only the admin can see verification dashboards
Visitor	Unregistered user. They can browse services, categories, and seller profiles without logging in, they cannot place orders. When trying to order, system asks them to register.

2.2. Requirement Identifying Technique:

The primary technique used to identify and structure the requirements of She Empower is Use-Case Analysis. It captures requirements from the perspective of actual users and maps every meaningful interaction between an actor and the system. The following use cases were identified through stakeholder discussions and analysis of the system scope.

Use Case 1: Register

Table 5 Use case 1: Register

Field	Description
Actor	Visitor / Buyer / Seller
Description	Users create a new account by entering name, email, phone, password, and Role (Buyer or Seller). Sellers also upload CNIC images and voice sample.
Pre-condition	User is not logged in.
Post-Condition	Accounts are created. Buyer becomes active. Seller account is marked as "pending verification".
Main Flow	1. User clicks "Register" and System shows registration form. 2. User enters name, email, phone, password, selects role. 3. If role is Seller, user uploads CNIC-image and voice sample. 4. User sends form. 5. System saves data on a Firebase and shows success message.
Alternate Flow	• If email already exists → error message. • If CNIC image invalid →, ask to re-upload.

Use Case 2: Login

Table 6 Use Case 2: Login

Field	Description
Actor	Registered Buyer / Seller / Admin
Description	User logs into the system using email and password.
Pre-condition	Users must be registered.
Post-Condition	User is redirected to role-based dashboard (Buyer, Seller, Admin)
Main Flow	1. User enters email and password. 2. System verifies credentials with Firebase Authentication. 3. System checks role and access granted – dashboard opens.
Alternate Flow	Wrong credentials → error message(Invalid email or password)

Use Case 3: Browse Services

Table 7 Use Case 3: Browse Services

Field	Description
Actor	Visitor, Buyer, Seller
Description	User browses categories without logging in or after login.
Pre-condition	None (Visitor can browse without account).
Post-Condition	Services are displayed in a list or grid.
Main Flow	1. User open home screen and system fetches services. 2. User clicks on a category and System displays that category.

Use Case 4: Place Order (Buyer)

Table 8 Use Case 4: Place Order (Buyer)

Field	Description
Actor	Buyer
Description	Buyer selects a service, fills a form and places their order.
Pre-condition	Buyer is logged in.
Post-Condition	Order saved in Firebase and seller receives notification.
Main Flow	1. Buyer selects a service and system shows order form. 2. Buyer fills in details and clicks "Place Order". 3. Buyer sees for confirmation message.
Alternate Flow	System asks to upload payment proof after order.

Use Case 5: Add New Skill (Seller)

Table 9 Use Case 5: Add New Skill (Seller)

Field	Description
Actor	Seller
Description	Seller adds a new skill/service (title, price, description)
Pre-condition	Seller is logged in and account is verified by admin.
Post-Condition	Skill is saved to Firebase and appears in seller's public profile.
Main Flow	1. Seller clicks "Add Skill" and system shows form. 2. Seller fills details and uploads images and system saves it. 3. Skills appear in seller's "My Skills" list.

Use Case 6: Update Order Status (Seller)

Table 10 Use Case 6: Update Order Status (Seller)

Field	Description
Actor	Seller
Description	Seller updates order status after serving the buyer.
Pre-condition	Seller has received an order (auto-accepted).
Post-Condition	Order status updated in database, buyer is notified.
Main Flow	1. Seller opens "Orders Received" screen and fetches orders. 2. Seller selects an order and chooses new status. 3. System updates status and buyer receives notification.

Use Case 7: Verify Seller (Admin)

Table 11 Use Case 7: Verify Seller (Admin)

Field	Description
Actor	Admin
Description	Admin reviews seller's CNIC image and voice sample.
Pre-condition	Seller has registered and uploaded documents.
Post-Condition	Seller's verification status updated (verified/rejected).
Main Flow	1. Admin opens "Payment Verifications" list. 2. Admin views screenshot, checks amount and order ID. 3. Admin clicks Verify or Reject and system updates payment status.

Use Case 8: Verify Payment (Admin)

Table 12 Use Case 8: Verify Payment (Admin)

Field	Description
Actor	Admin
Description	Admin reviews payment proof of buyer and marks as verified.
Pre-condition	Buyer has placed an order with manual payment method.
Post-Condition	Payment verified, order status may continue.
Main Flow	1. Admin opens "Payment Verifications" list and shows. 2. Admin verify screenshot and clicks "Verify" or "Reject". 3. System updates payment status and buyer notified.
Alternate Flow	Rejected – buyer can upload new proof.

Use Case 9: Chat Seller through WhatsApp:

Table 13 Use Case 9: Chat with Buyer/Seller

Field	Description
Actor	Buyer
Description	Buyer contacts the seller directly through WhatsApp regarding a service or order.
Pre-condition	Seller has a valid WhatsApp-enabled phone number.
Post-Condition	Communication continues through WhatsApp.
Main Flow	1. Buyer open seller profile and click Contact on WhatsApp. 2. Application opens WhatsApp conversation with the seller. 3. Buyer sends the message through WhatsApp.

Use Case 10: Browse as Visitor

Table 14 Use Case 10: Browse as Visitor

Field	Description
Actor	Visitor
Description	Unregistered users browse categories, and seller profiles.
Pre-condition	No login required.
Post-Condition	Visitors can view content but cannot place order.
Main Flow	1. Visitor opens app. 2. System shows top sellers, categories, services. 3. Visitor clicks on service to see details.
Alternate Flow	If visitor clicks Place Order, system shows "Please login".

Use Case 11: Manage Profile

Table 15 Use Case 11: Manage Profile

Field	Description
Actor	Buyer, Seller, Admin
Description	User updates profile information (name, phone, profile image)
Post-Condition	Profile updated in Firebase.
Main Flow	1. User goes to Profile screen. 2. User edits fields and saves. 3. System updates Firestore. and success message shown.
Alternate Flow	Invalid input (e.g., phone number wrong format) → error.

Use Case 12: Search Services

Table 16 Use Case 12: Search Services

Field	Description
Actor	Buyer, Visitor
Description	User searches services by keyword or category.
Pre-condition	None (search available for visitors too).
Post-Condition	Matching services show.
Main Flow	1. User type keyword in search bar. 2. System queries Firestore and results displayed in list.
Alternate Flow	No results → "No services found" message.

Use Case 13: View Order History

Table 17 Use Case 13: View Order History

Field	Description
Actor	Buyer
Description	Buyer views past and current orders with status.
Pre-condition	Buyer is logged in.
Post-Condition	Orders list displayed.
Main Flow	1. Buyer clicks "My Orders". 2. System fetches orders from Firestore (filter by buyerId). 3. Orders shown with status, date, total amount.

2.3 Functional Requirements

2.3.1 Functional Requirement FR-1

Table 18 Functional Requirement FR-1

Field	Value
Identifier	FR-1
Title	Voice-guided interface for local women
Requirement	The system shall support women in local areas by providing an easy interface through voice guidance on every screen.
Source	BR001
Rationale	Women with low digital literacy cannot read complex English instructions; spoken Urdu guidance makes navigation simple.
Business Rule	Voice guidance must be enabled by default; users can optionaly disable.
Dependencies	FR-3 (Urdu voice navigation)
Priority	High

2.3.2 Functional Requirement FR-2

Table 19 Functional Requirement FR-2

Field	Value
Identifier	FR-2
Title	Seller profile and WhatsApp contact
Requirement	The system shall allow sellers to create profiles containing their skills, sample images, and contact information. Buyers shall be

	provided with a WhatsApp button to contact the seller directly through WhatsApp.
Source	BR002
Rationale	Sellers need to display their services and communicate with potential buyers to complete orders.
Business Rule	Profile must include at least one skill image.
Dependencies	FR-7 (Chat system), FR-10 (Admin verification)
Priority	High

2.3.3 Functional Requirement FR-3

Table 20 Functional Requirement FR-3

Field	Value
Identifier	FR-3
Title	Urdu voice navigation (output only)
Requirement	The system shall provide Urdu voice-guided navigation where only the app speaks; users will not speak. The app provides voice output.
Source	BR003
Rationale	BR003
Business Rule	TTS must be Google Text-to-Speech with Urdu language set.
Dependencies	FR-1
Priority	High

2.3.4 Functional Requirement FR-4

Table 21 Functional Requirement FR-4

Field	Value
Identifier	FR-4
Title	Buyer browsing, ordering and WhatsApp contact
Requirement	System shall allow a buyer to browse categories, place orders using a order form, and communicate with the seller through WhatsApp.
Source	BR004
Rationale	Buyers need a unified, simple workflow from discovery to buy and post-order communication.
Business Rule	Order form must include delivery method and payment method.
Dependencies	FR-6, FR-7, FR-8
Priority	High

2.3.5 Functional Requirement FR-5

Table 22 Functional Requirement FR-5

Field	Value
Identifier	FR-5
Title	Auto order acceptance
Requirement	The system shall allow a seller to receive orders without manual acceptance or rejection (auto-acceptance).
Source	BR005

Rationale	To reduce seller workload and speed up order processing – especially for women with limited technical skills.
Business Rule	To reduce seller workload and speed up order processing – especially for women with limited technical skills.
Dependencies	FR-4
Priority	High

2.3.6 Functional Requirement FR-6

Table 23 Functional Requirement FR-6

Field	Value
Identifier	FR-6
Title	Delivery method choice
Requirement	The system shall allow a buyer to select the delivery method (self-pick or home delivery) in order form.
Source	BR006
Rationale	Buyers need flexibility to choose how they receive products (local pickup or home delivery).
Business Rule	Home delivery addresses must be provided if selected; self-pick does not require address.
Dependencies	FR-4
Priority	Medium

2.3.7 Functional Requirement FR-7

Table 24 Functional Requirement FR-7

Field	Value
Identifier	FR-7
Title	Payment method choice (COD / manual)
Requirement	The system shall allow a buyer to select the payment method: Cash-on-delivery or manual payment through Easypaisa or JazzCash in the same order form.
Source	BR007
Rationale	Manual payment (proof-based) is simpler for a college project and works with local mobile wallets.
Business Rule	If manual payment is selected, the buyer must upload a payment screenshot after order placement.
Dependencies	FR-4, FR-9
Priority	High

2.3.8 Functional Requirement FR-8

Table 25 Functional Requirement FR-8

Field	Value
Identifier	FR-8
Title	Order status update by seller
Requirement	The system shall allow a seller to update the order status to “nearby” or “delivered”.

Source	BR008
Rationale	Buyers need to know the real-time status of their orders; sellers can update after preparing/delivering.
Business Rule	Only the seller who received the order can update its status; buyer receives a notification on status change.
Dependencies	FR-5
Priority	Medium

2.3.9 Functional Requirement FR-9

Table 26 Functional Requirement FR-9

Field	Value
Identifier	FR-9
Title	Visitor browsing without account
Requirement	System shall allow a visitor to browse services without creating an account.
Source	BR009
Rationale	To attract potential users and allow them to explore before registration.
Business Rule	Visitors cannot place orders or chat; if they try, the system prompts registration.
Dependencies	None
Priority	Medium

2.3.10 Functional Requirement FR-10

Table 2-24 Functional Requirement FR-10

Field	Value
Identifier	FR-10
Title	Admin verification of payment receipts and sellers
Requirement	The system shall allow an admin to verify payment receipts and approve women sellers.
Source	BR010
Rationale	To prevent fraud and ensure only genuine women sellers (CNIC + voice) can run.
Business Rule	Admin can approve/reject seller registrations and payment proofs; notification sent to the user.
Dependencies	FR-2, FR-7
Priority	High

2.4 Non-Functional Requirements

2.4.1 Reliability

The system shall support stable operation during real-time chat and order updates. Meantime between failures (MTBF) shall exceed 24 hours under normal usage. Firebase offline persistence ensures data consistency when network is intermittent.

2.4.2 Usability

- Urdu voice guidance (Text-to-Speech) shall be provided on every screen to help users with low digital literacy.

- At least 80% of test users report that voice guidance is helpful.
- The interface shall follow Material Design guidelines with green/white themes, simple navigation, and consistent button placement.

2.4.3 Performance

- Application shall load within 3 seconds.
- Order placement, status update, and WhatsApp contact action shall complete within 2 seconds under normal 4G/5G Wi-Fi.
- Each panel (Seller, Buyer, Admin, Visitor) shall achieve 95% task completion accuracy (ordering, uploading, browsing)

2.4.4 Security

- User data (CNIC, number, email) shall be encrypted using Firebase Security Rules.
- Only authenticated users can read/write their own data; admin has special permissions for verification.
- Seller verification documents (CNIC image, voice sample) are stored in Firebase Storage with restricted access – only admin can view.
- No passwords are stored in plaintext, Firebase Authentication handles hashing.

2.5 External Interface Requirements

2.5.1 User Interfaces Requirements

- Urdu voice guidance (TTS) is available on every screen.
- Bottom navigation bar for buyers and sellers: home, explore, orders, profile.
- Admin dashboard has its own navigation: dashboard, users, verifications, payments, settings.
- Font size minimum 14sp, touch targets minimum 48dp.
- Color scheme: green (4CAF50) and white background.

- Layouts use ConstraintLayout and LinearLayout for responsiveness.
- Short feedback messages use Toast; important actions use Snackbar or dialog boxes.
- All text strings are stored in strings.xml for future Urdu localization.

2.5.2 Software interfaces

- Firebase Firestore (v34.12.0) – stores users, sellers, buyers, skills, orders, payments and verifications.
- Firebase Authentication (v34.12.0) – handles email/password registration and login.
- Firebase Storage (v34.12.0) – stores CNIC images, product images, payment proofs, voice samples.
- Google Text-to-Speech (Android API 21+) – provides Urdu voice output.
- Android OS (API 21+ / Android 5.0 Lollipop) – the platform.

2.5.3 Hardware interfaces

- Speaker – plays voice guidance (TTS).
- Microphone – used by sellers to record voice sample during registration.
- Camera (optional) – for taking CNIC photo or product images; gallery fallback available.
- Touchscreen – all user input (buttons, scrolling, forms).
- Internal storage – for Firebase offline persistence (caches profile and history).

2.5.4 Communications interfaces

- Wi-Fi or mobile data (4G/5G) is required for order placement, and database synchronization and opening WhatsApp for communication.
- Offline mode – when internet is unavailable, cached data is shown; write operations are queued and sync later.

2.6 Summary

Summary of Requirements Identification and Use Case Analysis for voice-guided marketplace app. Four main types of users were named, and each user type has distinctive features.The chapter found 24 functional requirements. It includes registration, profile management, ordering, payments, and admin controls. It also found 12 non-functional requirements. These cover reliability, usability, performance, and security. It described how the app will work with: Hardware parts and Software services like Firebase and Google TTS. This analysis helps in designing the system correctly.



Chapter 3: System Design

3. System Designs

The She Empower App built as a native Android application designed to connect and help women sellers with local buyers through voice-guidance interface. This app will be built using Java in Android Studio for the frontend and firebase for backend. This application will use the T.T.S (text to speech) library for Urdu voice-guidance and secure proof-based manual payment system.

3.1 Design Considerations:

This section outlines the assumptions, dependencies, limitations, and risks that have shaped the design of the She Empower system:

1. Assumptions:

- Users will have basic knowledge of using smart phones such as open apps and tap buttons.
- Users don't try to use fake voices in verification methods.
- The Internet is needed for real-time chat features.

2. Dependencies:

- **Frontend (Android Java):** For activities, fragments, layouts, product listings, and order management.
- **Backend (Firebase): Manages** Authentication, Firebase, storage of images, payment proof images and real-time chat messages.
- **Voice service (Google TTS):** Urdu guidance for navigation and instructions.
- **Payment verification system:** Manages proof-based transactions through image upload and admin verification.

3. Limitations:

- Initially the app was designed for local areas and not tested for other cities.
- Payment Proofs are verified by the admin. This can cause delays in order confirmation.
- Without the Internet, users can't send or receive messages or get live order status.

4. Risks & Mitigations:

Table 27 Risks and mitigation

Risks	Impact	Mitigation
Data Security	Risk of personal information theft	Use Firebase security rules and end-to-end encryption.
User Adoption	Low user engagement due to low digital literacy.	Continuous improvement through user feedback, voice-guided tutorials, and marketing.
Payment Disputes	Fraud and disputes.	Admin checks payment proof.

3.2 Design Models:

Provide the descriptions of the following models used to describe the system design. Also, ensure visibility of all diagrams. The applicable models for the project using an object-oriented development approach may include:

3.2.1 UML Class Diagram

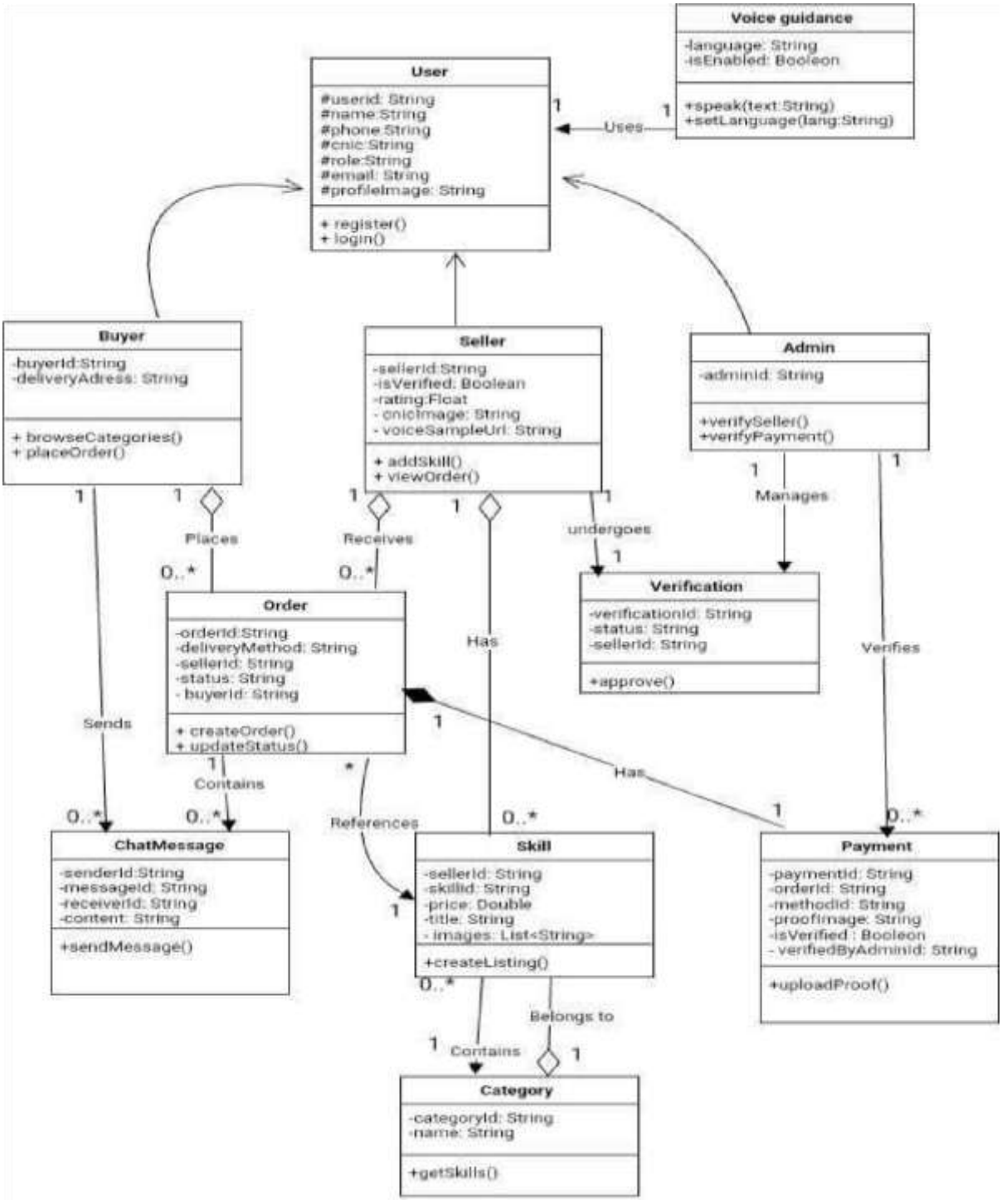


Figure 1 Class diagram

Description:

This diagram is the static structure by showing its attributes and relations between them.

1. **User (Parent Class):** The User class stores basic information like user ID, name, phone, CNIC, role, email, and profile image. It provides registration and login methods.

2. **Voice Guidance:** This class manages Urdu voice output. It stores language (Urdu) and provides methods to speak text and change language settings using Google TTS.

3. **Buyer (Child):** The Buyer class inherits from User and is customers who browse services, place orders, and manage their purchases. It includes buyer ID and delivery Address attributes and provides methods to browse categories and place orders.

4. **Seller (Child):** The Seller class inherits from user and is women who sell their skills. It has a seller ID, verification status, rating, CNIC image, and voice sample URL. It can browse categories and place orders.

5. **Admin (Child):** The admin class manages the system and verify sellers and payments.

6. **Order:** This class stores order details like order ID, delivery method, seller ID, status, and buyer ID. It can create an order and update its status.

7. **Category:** This class represents skills and control skills under that category.

8. **Payment:** This class manages payment transactions. It stores paymentId, orderId, method, proof image, verification status, and the admin who verified it.

9. **Skill:** This class represents a service or product offered by a seller. It stores skillId, sellerId, categoryId, title, description, price, and images.

10. **Verifications:** This class manages seller verification. It has a verification ID, status, and seller ID. It can approve a seller.

3.2.2 State Transition Diagram

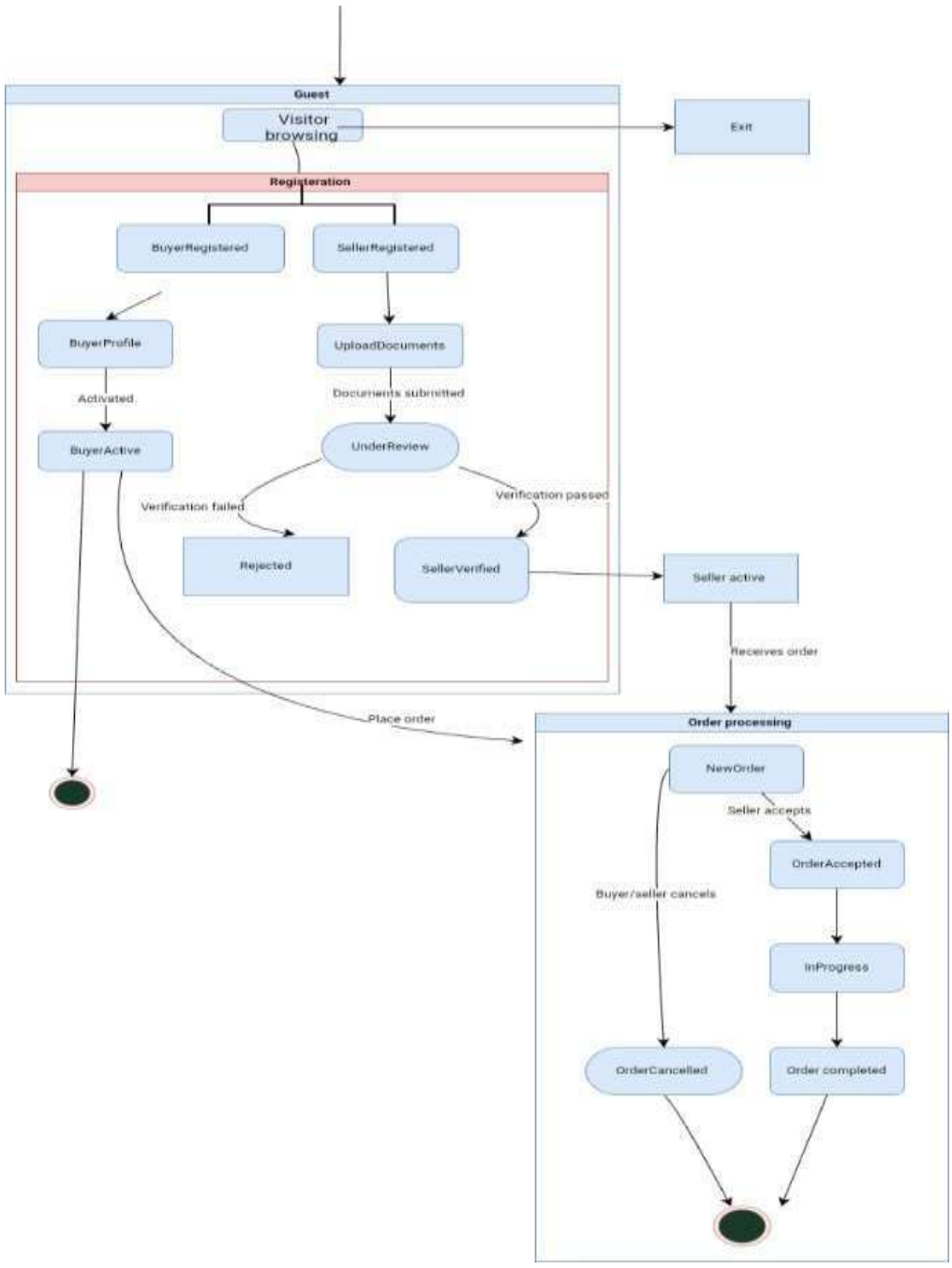


Figure 2 State transition Diagram

Description:

The detailed description of the state transition diagram for this app is as follows:

States:

1. **Initial State _Visitor browsing:** The user starts as a guest who can browse services, view categories and seller profiles without logging in. To place an order or sell services, the user must log in.

2. **Registration:** The user registers as a buyer or seller. Buyer registration requires basic information.

3. Seller registration: It requires CNIC and voice sample.

3. Registration Attempted: A new user completes the registration form.

4. Entering Documents (for Seller): The seller inputs CNIC image and voice sample.

5. Document Validation: Once documents are submitted, they divided into two sections:

- **Invalid Documents:** System rejects the seller and goes to Rejected State.
- **Invalid Documents:** System rejects the seller and goes to Rejected State.

6. Buyer (if role is buyer)

Buyer Registered → Buyer Active → Place order → New Order → Order Accepted → InProgress → Order completed → Order Cancelled (if buyer cancels) → End

7. Seller (if role is seller):

Seller Registered → Upload Documents → Under Review → Verification passed – Seller Verified → Verification failed – Rejected → Activated – Seller Verified → Place order → New Order → Seller accepts – Order Accepted → In Progress → Order completed → Order Cancelled (if seller cancels) → End

3.3 Architectural Design

3.3.1 UML Component diagram

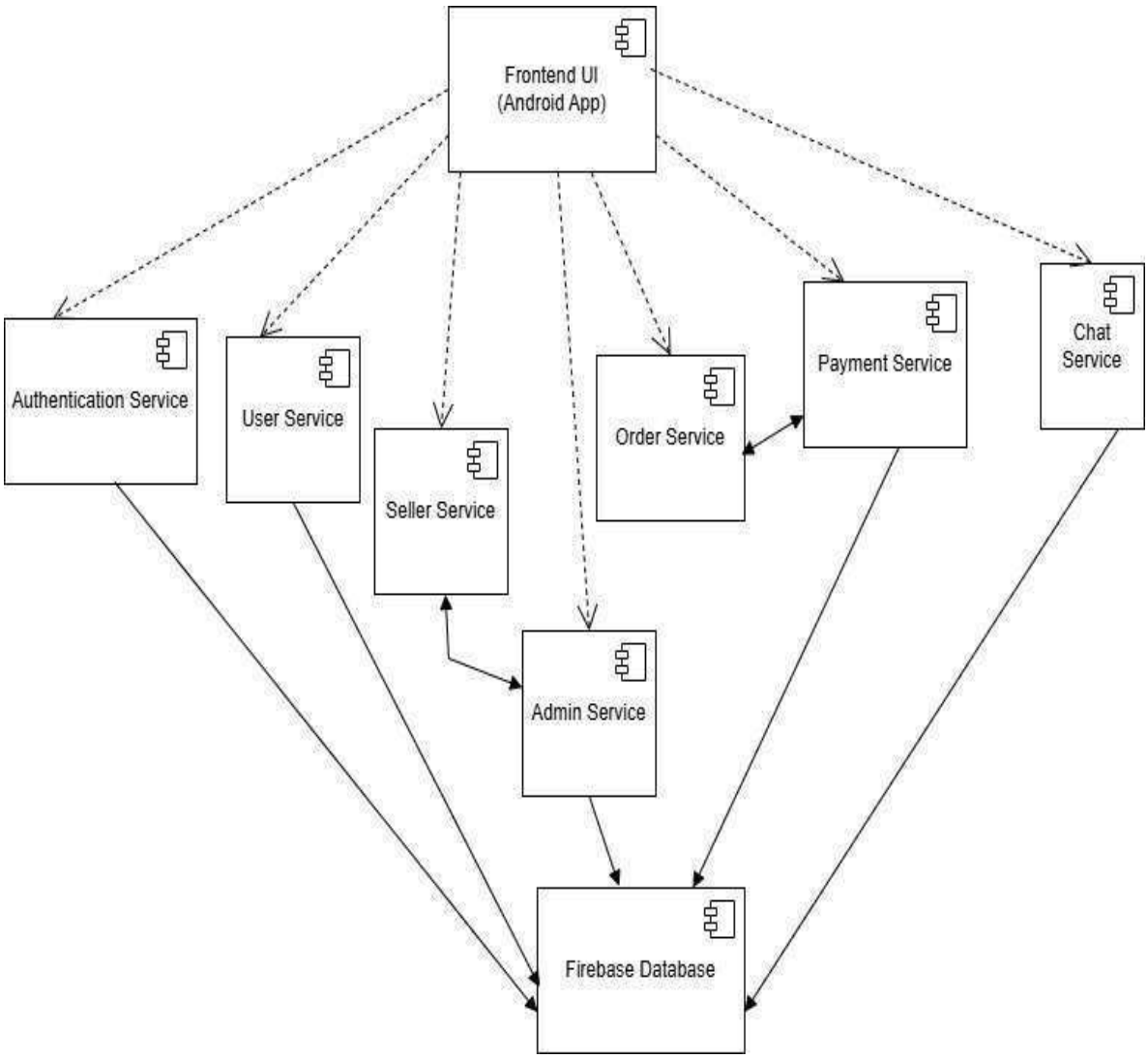


Figure 3 UML Component Diagram

Description:

1. **Frontend UI:** The frontend UI is a user-facing component with which visitors, buyers, sellers, and admins interact with the application. It controls UI elements like browsing service, voice guidance, and order forms of ad chat screens. It receives input and gives output by sending requests to the backend.
2. **Authentication Service:** The authentication service ensures that only authenticated users can access the application resources. It manages the login operation, registration process, and user credentials.
3. **User Service:** User service manages profile-related operations like creation of buyer and seller profiles, updates of user information, and store CNIC image and voice sample. It interacts with databases to store and change data.
4. **Order Service:** The order service manages transaction processes, order placement by the buyer, auto acceptance by the seller, and order status update. It also links the order to buyers, sellers, and services.
5. **Payment Service:** Payment service manages manual transaction processes and stores payment receipts uploaded by buyers, allows admin to verify and update payment status.
6. **Admin Service:** Admin Service manages seller verification, payment receipt verification, user management, and report generation. Only the admin can access these services.
7. **Seller Service:** Manages seller-specific features like skill management and order viewing.
8. **Chat Service:** Manages real-time messaging between buyers and sellers.
9. **Database:** Database stores all data of the application. It stores data from user accounts, seller profiles, buyer order history, services, orders, chats, payment history, and

verification requests. The application services interact with the database to update, insert, or remove data.

3.3.2 MVC Layer Architecture

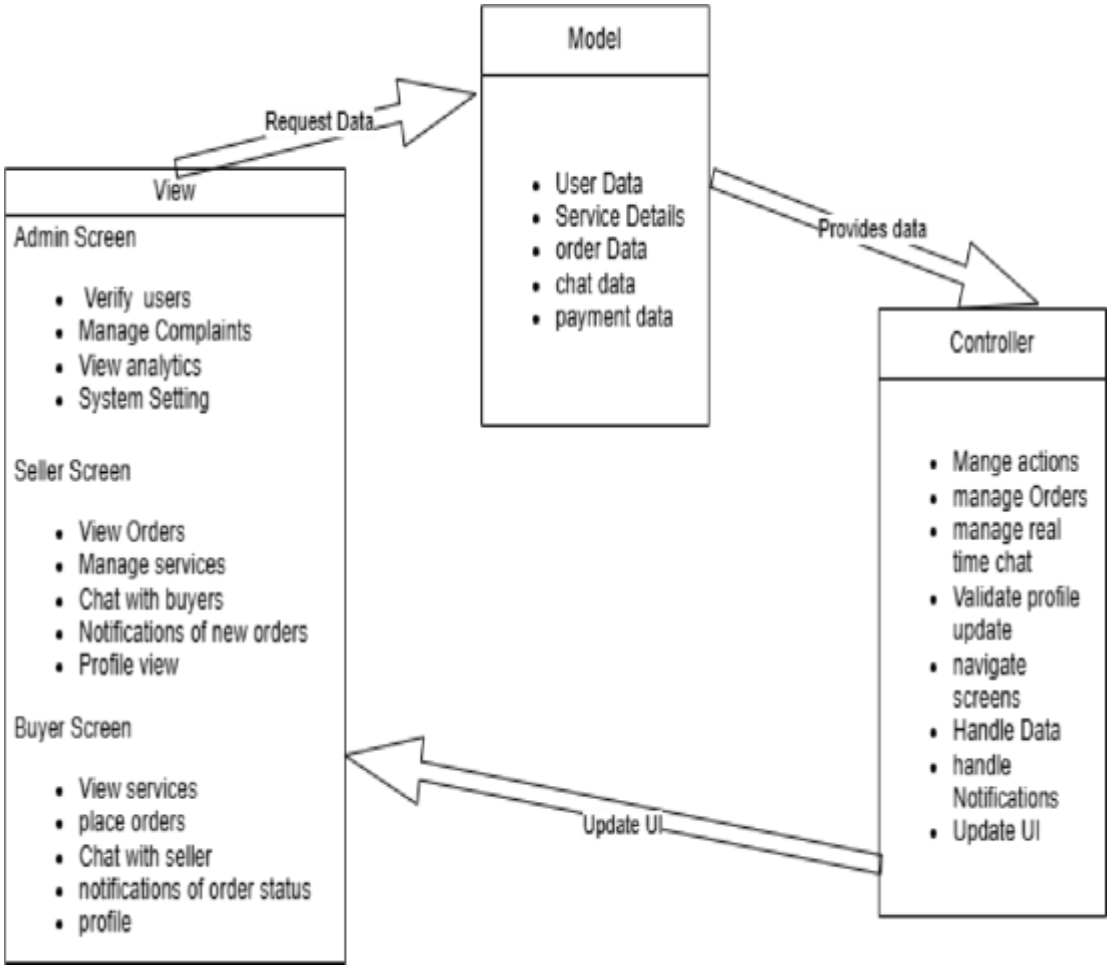


Figure 4 MVC Layer Architecture Description:

Description:

The description for the MVC model is as follows.

1. View Layer: There are three screens.

The admin screen verify users, manage complaints, and change settings.

Seller screen view orders, manage services, update order status and contact through WhatsApp.

Buyer screen browse services, place orders, contact sellers through Whatsapp and view order status updates.

2. **Model Layer:** This holds all the app data. User Data stores profiles and login info. Service Details stores product info. Order Data keeps transaction records. Payment data stores receipts and verification details.
3. **Controller Layer:** This manages all the logic between View and Model. It manages orders, WhatsApp contact ,external application launch and confirms profile updates, moves between screens, processes data, sends notifications, and updates to the UI.

3.4 Data Design

This section is about how the She Empower system takes all the information and turns it into data. The system uses an Object-Oriented approach. So the data design is presented in terms of objects their attributes and the things they can do.

3.4.1 Data Dictionary

Here is a list of all the things in the She Empower system in alphabetical order along with what type of data they are and what they mean:

Table 28 System Entities

Entity	Description
Admin	The person who manages the system checks sellers and payments.
Buyer	A person who looks at services, places orders, manages their purchase.
Category	The groups of skills like Cooking or Stitching.
order	The record of a transaction between a buyer and a seller.
Payment	The details of a transaction, including how it was paid and how much.
Seller	A woman who sells things on the system, lists her skills. Manages her orders.
Skill	A Product that a seller offers.
User	The information about a person who uses system, which includes all types of users.
Verification	The process of checking a sellers documents and identity including checking their CNIC and voice.
VoiceGuidance	The system that helps people navigate the She Empower system by reading out text in Urdu.

Entities with Attributes

1. **User**

    - **userId (String):** Unique identifier for the user.
    - **name (String):** Full name of the user.
    - **phone (String):** Contact phone number.
    - **cnic (String):** CNIC number for identity.
    - **role (String):** Role of the user – Buyer, Seller, or Admin.
    - **email (String):** Email address for login.

2. **VoiceGuidance**

    - **language (String):** Language used for speech (default "Urdu").
    - **isEnabled (Boolean):** Whether voice guidance is on/off.

3. **Buyer**

    - **buyerId (String):** Unique buyer identifier.
    - **deliveryAddress (String):** Default address for orders.

4. **Seller**

    - **sellerId (String):** Unique seller identifier.
    - **isVerified (Boolean):** Whether the seller is approved by admin.
    - **cnicImage (String):** URL of uploaded CNIC image.
    - **voiceSampleUrl (String):** URL of voice sample for verification.

5. **Admin**

    - **adminId (String):** Unique admin identifier.

6. **Order**

- **orderId (String):** Unique order identifier.
- **deliveryMethod (String):** Self-pick or home delivery.
- **sellerId (String):** ID of the seller who receives the order.
- **status (String):** Order status (e.g., Accepted, In Progress, Completed)
- **buyerId (String):** ID of the buyer who placed the order.

7. **SendMessage**

- **messageId (String):** Unique message identifier..
- **timestamp (DateTime):** Time the message was sent.

8. **Verification**

- **verificationId (String):** Unique verification record ID.
- **status (String):** Pending, Approved, or Rejected.
- **sellerId (String):** ID of the seller being verified.

9. **Payment**

- **paymentId (String):** Unique payment identifier.
- **methodId (String):** Payment method (COD, Easypaisa, JazzCash).
- **proofImage (String):** URL of uploaded payment screenshot.
- **isVerified (Boolean):** Whether admin verified the payment.
- **verifiedByAdminId (String):** ID of admin who verified.

10. **Skill**

- **skillId (String):** Unique skill identifier.
- **sellerId (String):** ID of seller offering the skill.
- **price (Double):** Price of the service.

- **title (String):** Name of the skill.
- **images (List\<String\>):** List of sample image URLs.

11. Category

- **categoryId (String):** Unique category identifier.
- **name (String):** Category name (e.g., Cooking, Stitching)

Methods

Table 29 Methods

Entities	Methods	Description
User	register()	Creates a new user account
	login()	Authenticates user with email.
Voice-Guidance	speak(text)	Converts text to Urdu speech.
	setLanguage(lang)	Changes the language of voice output.
Buyer	browseCategories()	Views all product categories.
	placeOrder()	Creates a new order.
Seller	browseCategories()	Views categories.
	placeOrder()	(used in diagram, same as buyer).
	createListing()	Adds a new skill/service.
Admin	verifySeller()	Approves or rejects seller verification.
	verifyPayment()	Approves or rejects payment proof.

Order	createOrder()	Saves a new order to database.
	updateStatus()	Changes order status.
SendMessage	sendMessage()	Saves and delivers message to receiver.
Verification	approve()	Marks seller verification as approved.
Payment	uploadProof()	Uploads payment screenshot image.
Skill	createListing()	Adds a new skill.
Category	getSkills()	Fetches all skills under a category.

Methods with Parameters

1. **User**

- register(name, email, password, phone, role)
- login(email, password)

2. **VoiceGuidance**

- speak(text, language)
- setLanguage(lang, isEnabled)

3. **Buyer**

- browseCategories(categoryId)
- placeOrder(buyerId, sellerId, skillId, deliveryMethod, totalAmount)

4. **Seller**

- browseCategories(categoryId)
- placeOrder(sellerId, buyerId, skillId, deliveryMethod, totalAmount)
- createListing(sellerId, title, price, description, images, categoryId)

5. **Admin**

- verifySeller(sellerId, status, adminId)
- verifyPayment(paymentId, isVerified, adminId)

6. **Order**

- createOrder(buyerId, sellerId, skillId, deliveryMethod, totalAmount)
- updateStatus(orderId, newStatus)

7. **ChatMessage**

- sendMessage(senderId, receiverId)

8. **Verification**

- approve(verificationId, adminId, status)

9. **Payment**

- uploadProof(orderId, methodId, proofImageUrl)

10. Skill

- createListing(sellerId, title, price, description, images, categoryId)

11. Category

- getSkills(categoryId)

3.5 User Interface Design

This section provides an overview of the She Empower user interface from the user's perspective. The interface follows a clean, consistent design language across all pages.

Detailed wireframes and prototypes are provided in Appendix

3.5.1 Screen Images

The following screenshots illustrate the key screens of the She Empower platform:

1. Login Screen



Figure 5 Login Screen

2. **Seller Registration**



Figure 6 Seller Registration

3. Visitor Screen

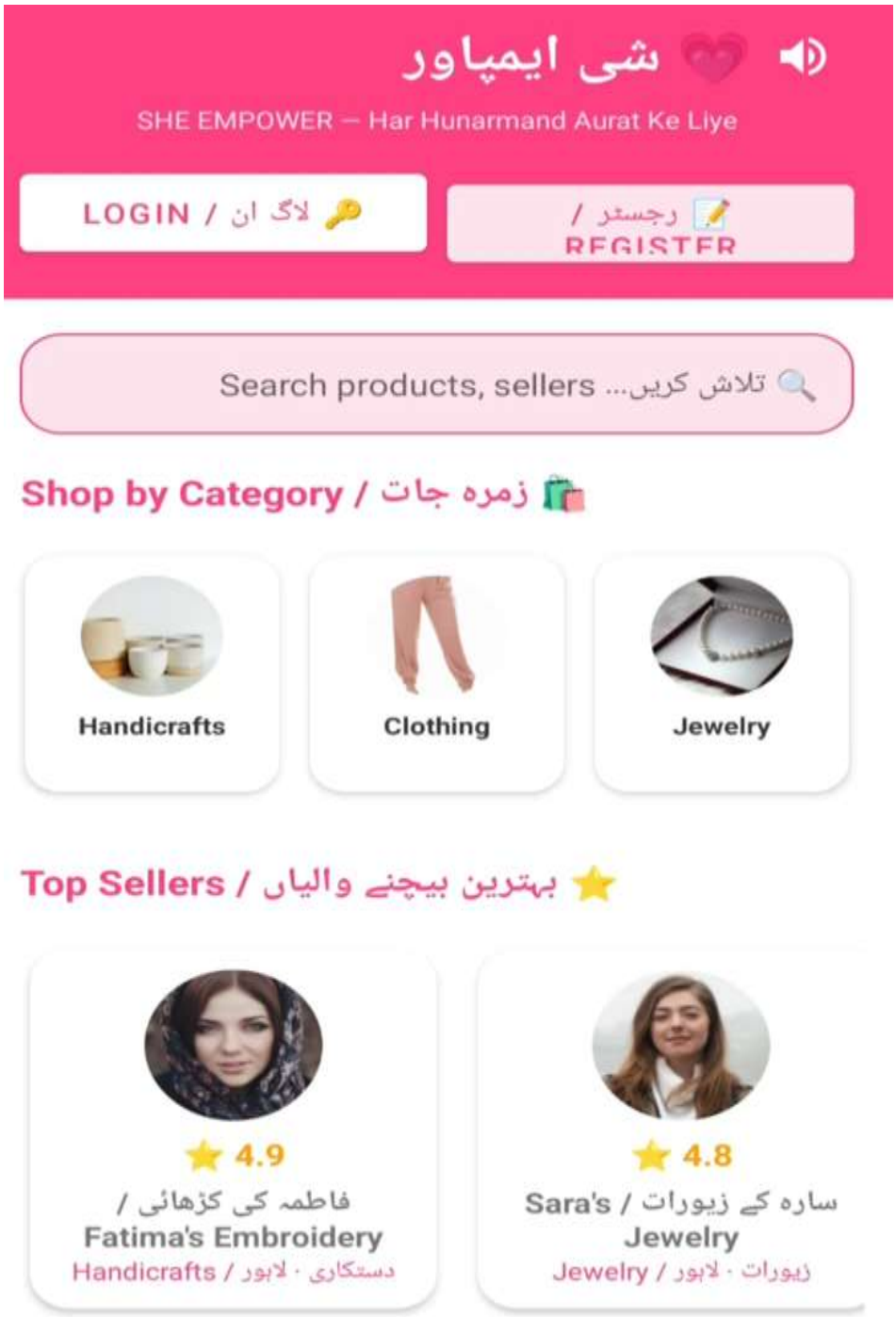


Figure 7 Visitor Screen

4. Admin Screen

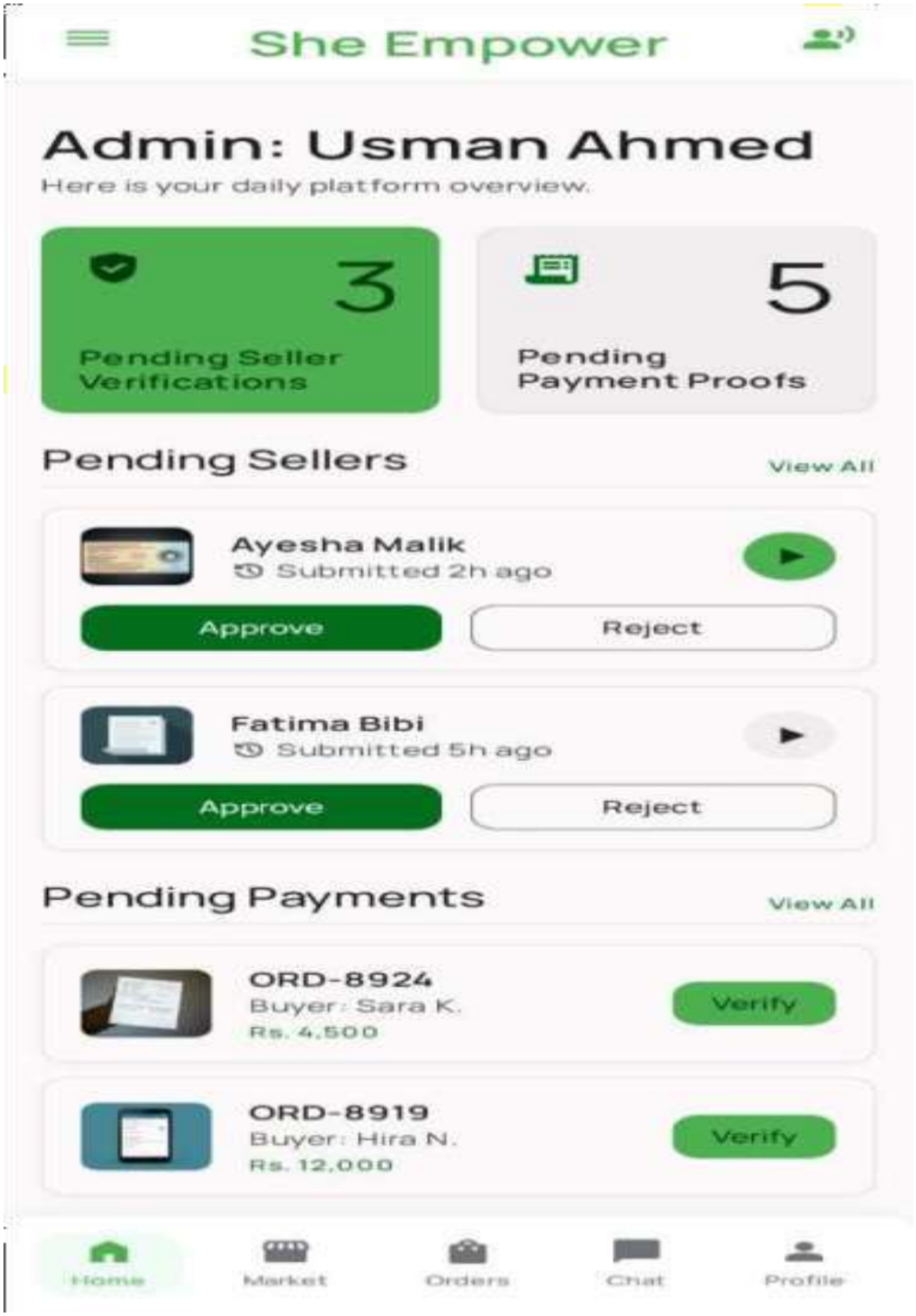


Figure 8 Admin Screen

3.5.2 Screen Objects and Actions

The following tables describe the screen objects and their actions for two key use cases.

1. Seller Registration Screen

Screen Object	Action
Name Input Field	Type seller's full name
Email Input Field	Type email address
Phone Input Field	Type phone number
Password Input Field	Type password
Upload CNIC Button	Opens camera/gallery to capture CNIC image
Record Voice Button	Starts recording a voice sample for verification
Register Button	Submits all details; creates account with pending verification

2. Unified Order Form (Buyer)

Screen Object	Action
Skill Title & Price	Display selected skill details
Delivery Method	Select delivery options
Payment Method (COD/JazzCash)	Select payment method
Upload Payment Proof Button	Opens gallery to attach payment screenshot (visible when online method is selected)
Submit Order Button	Validates form and places order; shows confirmation toast

3.6 Behavioral Model:

3.6.1 Use Case Diagram

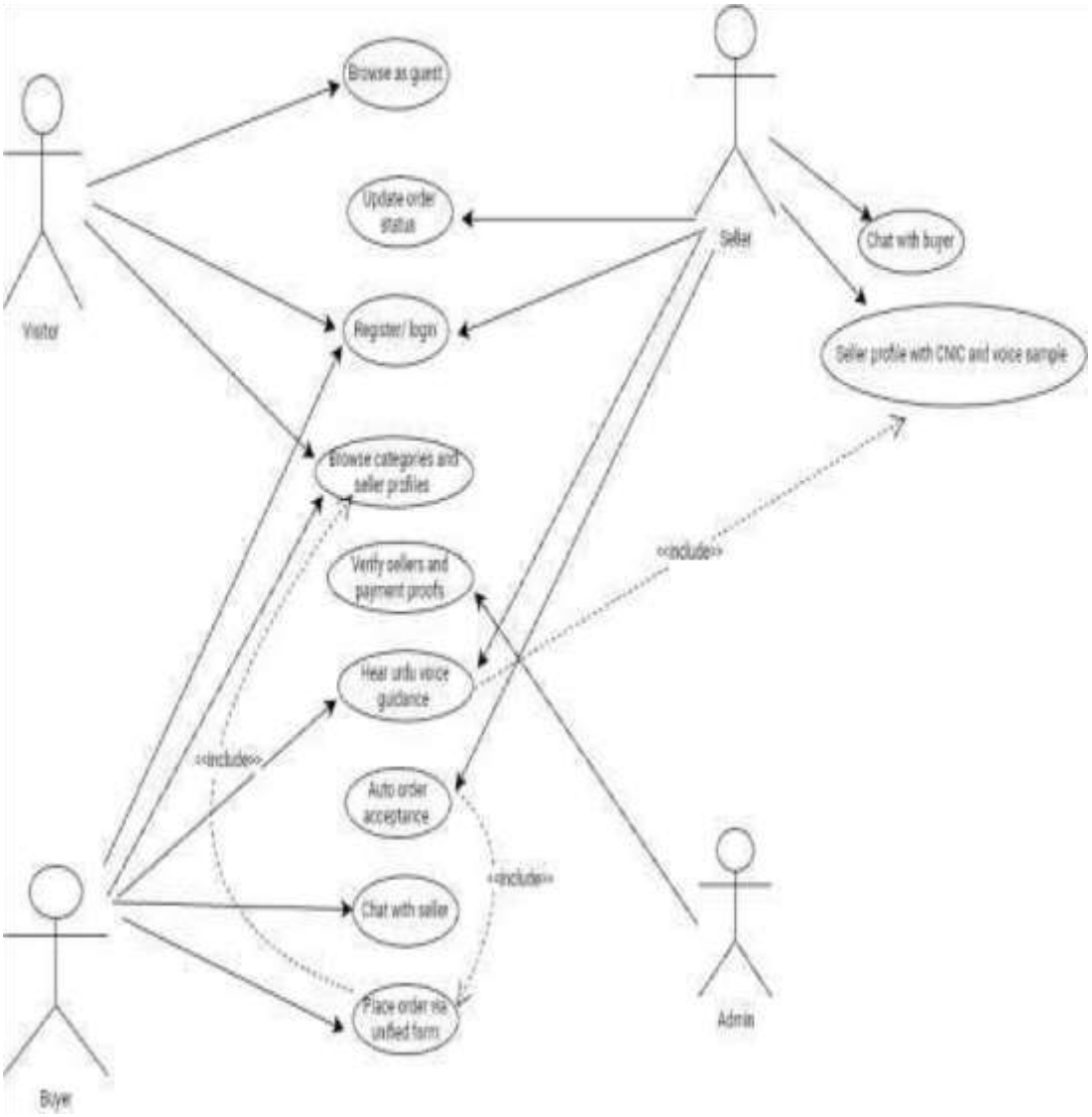


Figure 9 Use Case Diagram

Description

UseCase-1 (Browse Services): A visitor or guest can browse available services on the platform without creating an account or logging in.

UseCase-2 (View Seller Products): A visitor can view detailed seller products, such as images and videos uploaded by the seller, etc.

UseCase-3 (View Top Sellers): A visitor can view a list of top-rated sellers featured on the platform.

UseCase-4 (Add New Services): A seller can add new skill-based services to their profile for buyers.

UseCase-5 (Receive Orders): A seller can receive orders automatically from buyers without manual acceptance or rejection.

UseCase-6 (Update Order Status): A seller can update the status of an order, such as "Nearby", "In Progress", or "Delivered".

UseCase-7 (Contact Seller): A Buyer can communicate with the Seller through WhatsApp.

UseCase-8 (Verify Sellers): An admin can verify seller registrations by checking CNIC images, voice samples, and payment proofs.

UseCase-9 (Manage Users): An admin can manage all registered users, including buyers, sellers, and visitors.

UseCase-10 (Monitor System): An admin can check overall system performance and user activity.

UseCase-11 (Register/Login): A visitor, seller, or buyer can register a new account or log into an existing account to access their dashboards.

UseCase-12 (Search Services): A buyer can search for specific services or skills using keywords or category filters.

UseCase-13 (Place Order): A buyer can place an order for a selected service using the unified single order form.

3.6.2 Interaction Diagram (sequence/collaboration)

1. Voice-Guidance UI Sequence Diagram

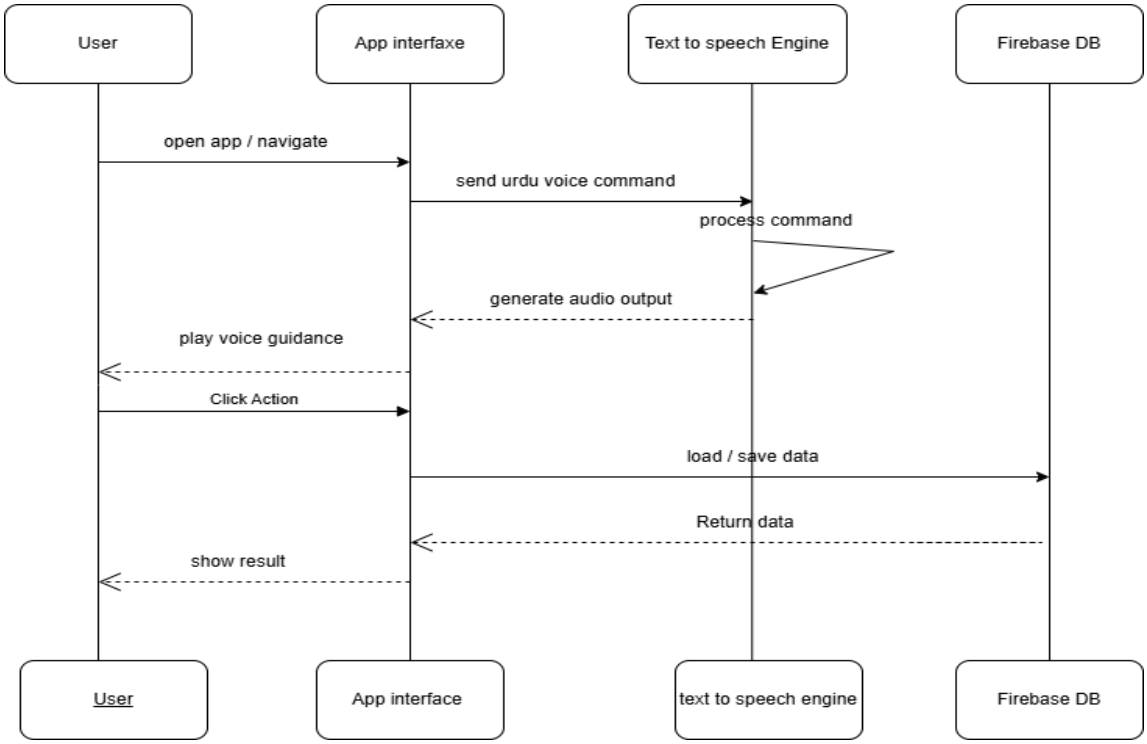


Figure 10 Voice-Guidance UI Sequence Diagram 1

Description:

- User opens the app or navigates to any screen.
- Interface sends an Urdu text command to the Text-to-Speech engine.
- The TTS engine processes the command and generates audio output.
- Audio is played to the user as voice guidance.
- User performs the next action (clicks on the button).
- Interface loads or saves data to the Firebase database.
- The result is then displayed to the user.

2. Seller Profile UI Sequence Diagram

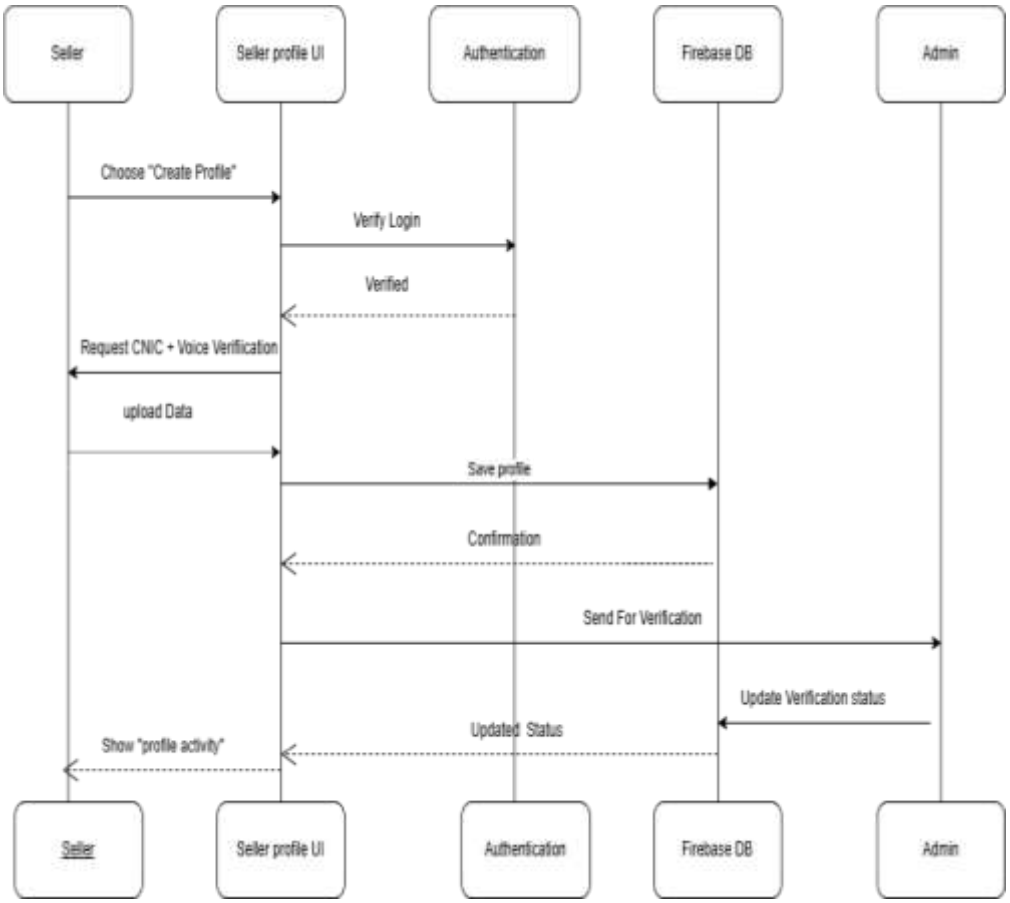


Figure 11 Seller Profile UI Sequence Diagram 2

Description:

- Seller selects the "Create Profile" option from the interface.
- App checks authentication and asks for CNIC and voice verification data.
- Seller uploads CNIC images and voice samples.
- The interface sends this data to the Firebase database.
- Data is saved and sent to the admin for verification.
- Admin verifies documents and updates the verification status.
- The status is saved in the database and shown to the seller.

3. Category Lists UI Sequence Diagram

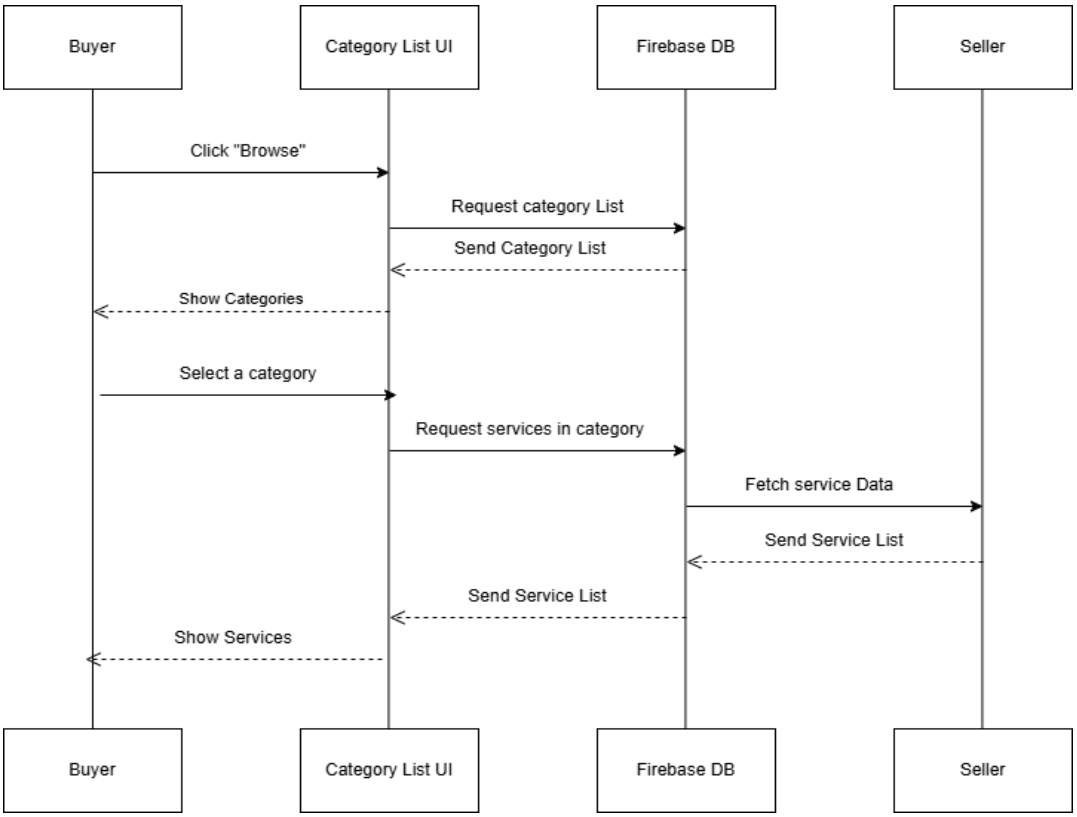


Figure 12 Category Lists UI Sequence Diagram 3

Description:

- The buyer clicks the "Browse" button.
- Category UI requests the category list from the Firebase database.
- The database then returns to the list of categories of UI.
- Categories are displayed to the buyer.
- Buyer selects a category.
- UI requests services in that category from Firebase.
- The database fetches service data from sellers and returns it to the service list.
- Services are displayed to the buyer.

4. Unified Buyer Form UI Sequence Diagram

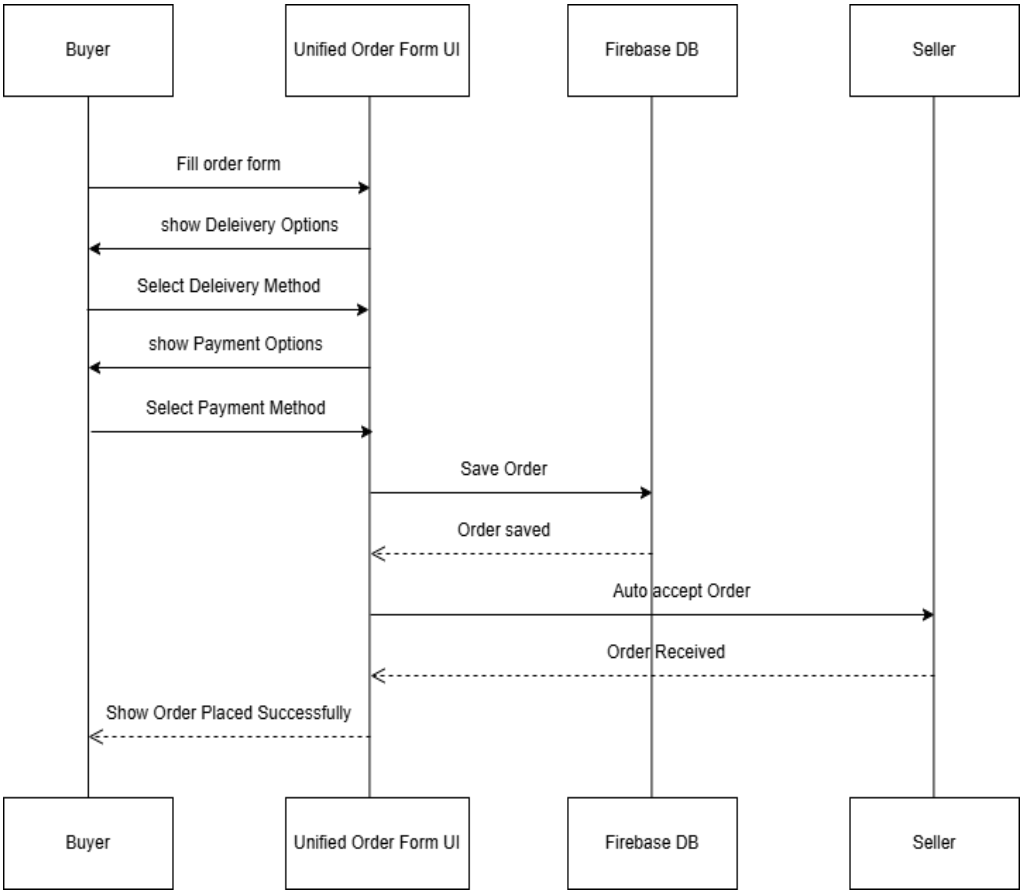


Figure 13 Unified Buyer Form UI Sequence Diagram 4

Description:

- The buyer fills out the order form on the interface.
- The form shows delivery and payment options with voice guidance.
- Buyer selects delivery method and payment choice.
- The form saves the order in the Firebase database.
- The order is auto accepted and sent to the seller's dashboard.
- Buyer receives an "Order Placed Successfully" confirmation message.

5. Auto Order Acceptance

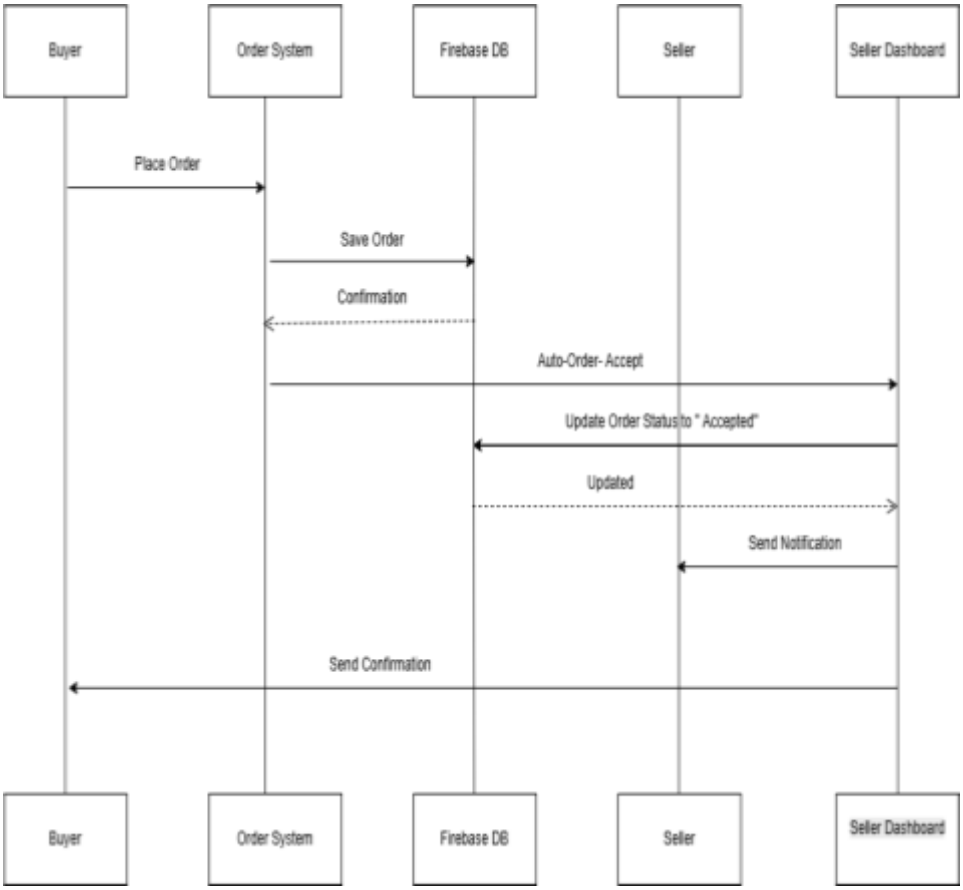


Figure 14 Auto Order Acceptance 5

Description:

- Buyer places an order through the interface.
- The order system saves the order in the Firebase database.
- Once confirmed, the system auto-accepts the order.
- The order is sent to the seller's dashboard.
- Dashboard updates the order status to "Accepted" in the database.
- Seller is notified, and confirmation is sent to the buyer.

6. Order Status UI Sequence Diagram

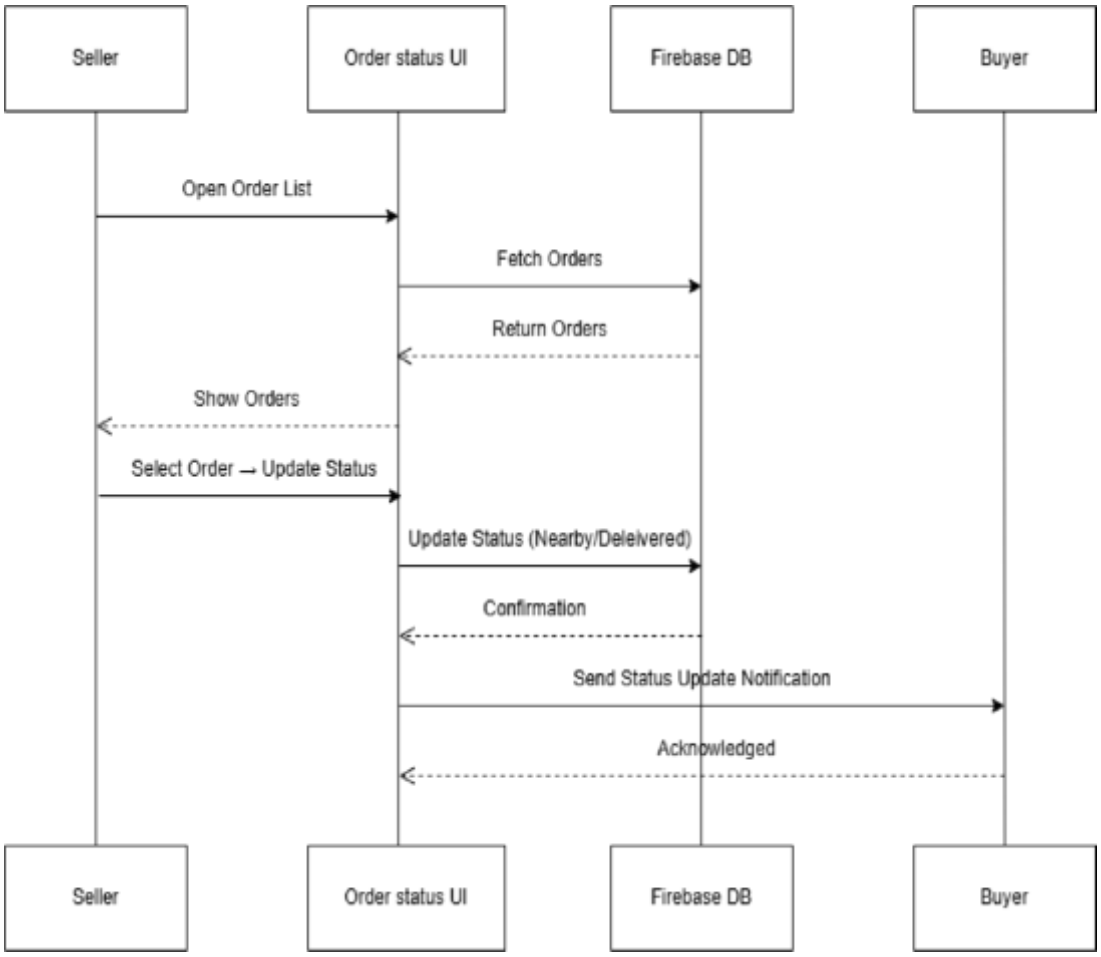


Figure 15 Order Status UI Sequence Diagram 6

Description:

- Seller opens the order list from the Order Status UI.
- UI fetches orders from the Firebase database.
- The orders are displayed to the seller.
- Seller selects an order and updates its status.
- UI updates this status in the Firebase database.
- A notification is sent to the buyer about the status change.

7. Visitor UI Screen Sequence Diagram

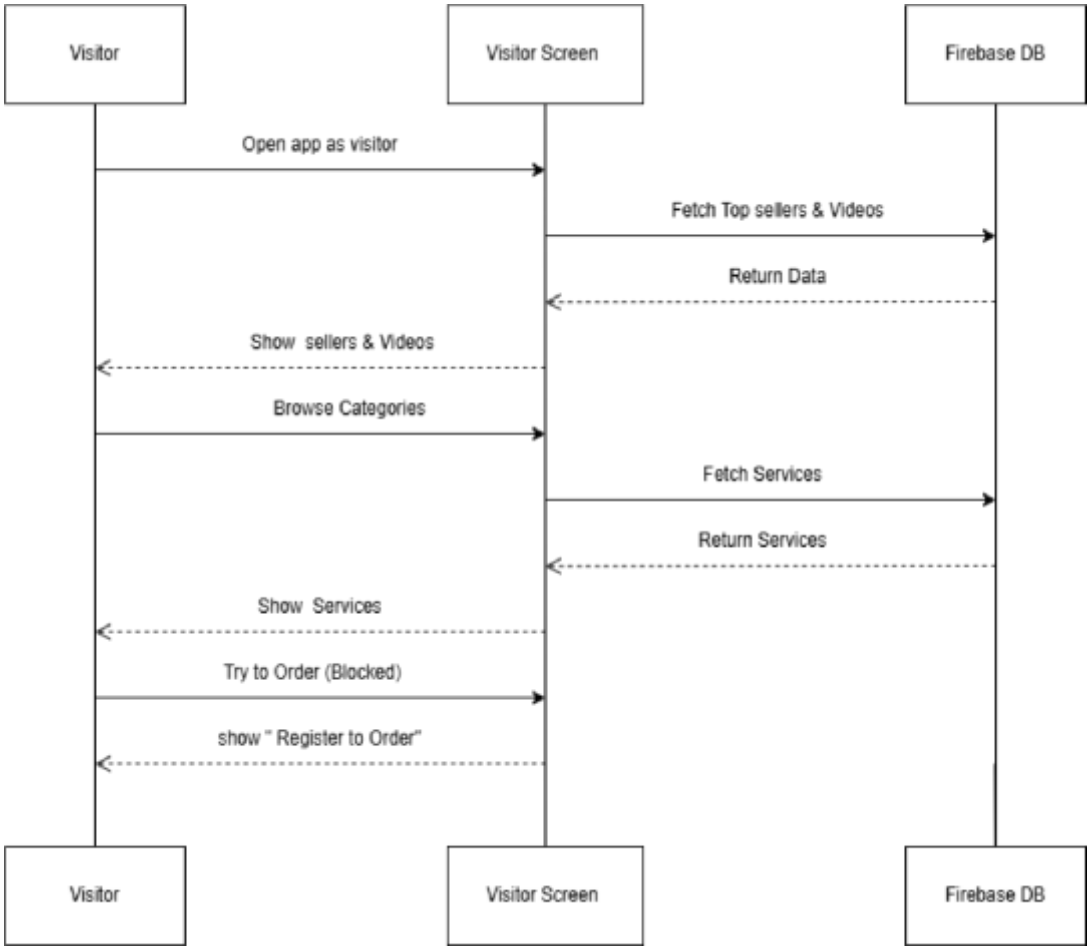


Figure 16 Visitor UI Screen Sequence Diagram 7

Description:

- Visitors open the app as guests.
- Visitor UI fetches top sellers and service videos from Firebase.
- Data is displayed to the visitor.
- Visitors can browse categories and services freely.
- If a Visitor tries to place an order, they are blocked.
- App prompts the visitor to register first.

8. Admin Dashboard UI Sequence Diagram

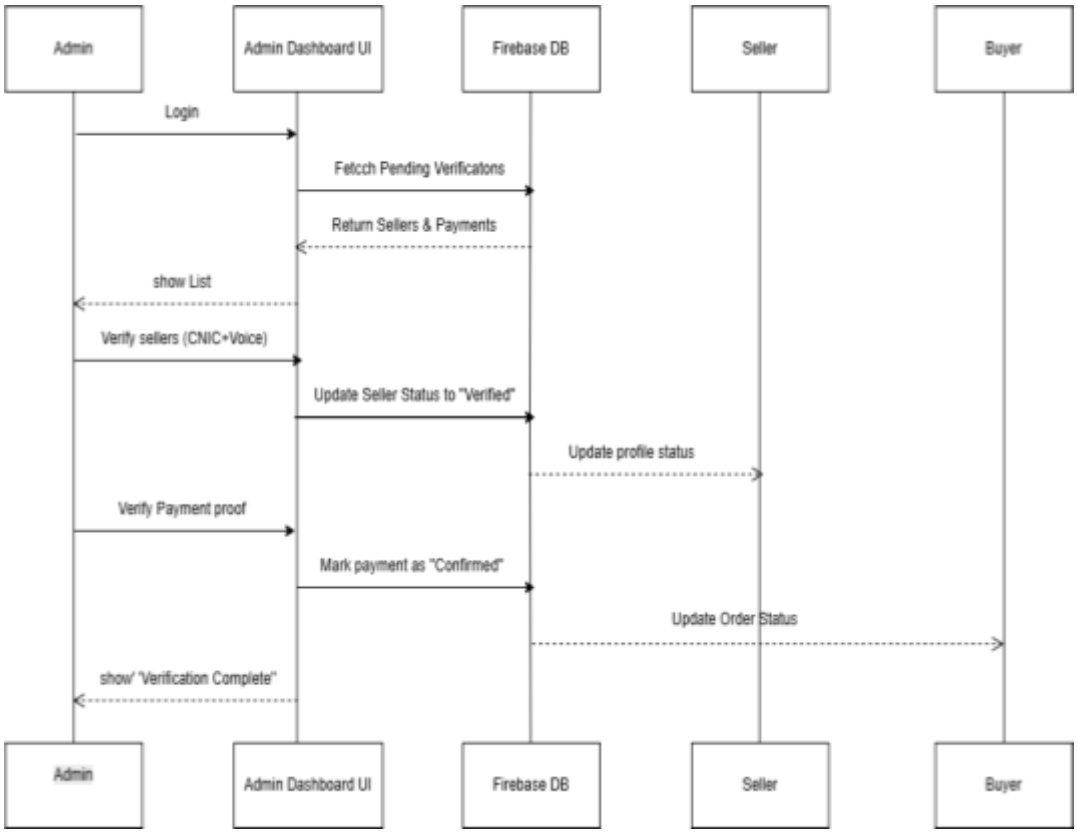


Figure 17 Admin Dashboard UI Sequence Diagram 8

Description:

- Admin logs into the dashboard.
- Dashboard fetches pending verifications (sellers and payments) from Firebase.
- The pending requests are displayed to the admin.
- Admin verifies a seller by checking CNIC and voice data.
- Admin confirms payment by reviewing the payment proof.
- After verification, the database is updated.
- Notifications are sent to both the seller and the buyer.

3.7 Design Decisions

The design choices while developing the She Empower app are:

Design Choice

1. **Frontend Framework**

Android Native (Java): We chose Android Studio with Java to build the user interface. Android Native provides better device integration (camera, microphone), offline support, and performance for mobile devices.

2. **Backend Technology**

Firebase: We use Firebase as the backend platform. It provides authentication, a Firestore database, cloud storage for application data. Communication between buyers and sellers is handled externally through WhatsApp.

3. **Database Management System**

Firebase Firestore: We selected Firestore as the NoSQL database. It supports realtime data sync, offline persistence, and a flexible schema, which is useful for storing users, orders, services.

4. **Authentication and Authorization**

Firebase Authentication: We use Firebase Email/Password authentication. Rolebased access (Buyer, Seller, Admin, Visitor) is implemented using a userType field in the database.

5. **Voice Guidance Service**

Google Text-to-Speech (Urdu): We integrated the Google TTS library for Urdu voice guidance. The app speaks instructions to help women users navigate easily.

Only text-to-speech is used; speech recognition is not needed.

6. **Payment Verification System**

Manual Proof-based Verification: We chose manual verification instead of automated payment gateways. Buyers upload payment screenshots, and the admin verifies them. This is simpler and works with local methods like Easypaisa and JazzCash.

7. **API Design**

Firebase SDK Direct Calls: We do not create separate REST APIs. The Android app calls Firebase SDK directly for database operations, authentication, and storage. This reduces development time.

8. **Error Handling and Logging**

Try-Catch Blocks: We used try-catch blocks to manage runtime errors such as network failures or null data. Firebase Crashlytics is used to track crashes.

9. **Code Organization and Modularity**

MVC Pattern: We organized code using the Model-View-Controller pattern. Activities and Fragments (View), View Models (business logic), and Repositories (control layer) are kept in separate folders following a consistent naming convention.

10. **Offline Support**

Firebase Offline Persistence: It allows users to access cached profile and order information when internet connectivity is temporarily unavailable.

11. Minimum Device Support

API Level 21+ (Android 5.0 Lollipop): We set the minimum API level 21 to support lowend devices. Voice guidance works on all supported versions.

3.8 Summary

This chapter explains the software design for ‘She Empower’ app. It listed assumptions, risks, and constraints for building app. Class diagrams showed how Users, Sellers, Buyers, and Orders connect. State diagrams showed seller verification steps. Sequence diagrams showed voice help, order placement, and chat. Component and MVC diagrams showed how UI, Firebase, and logic are separate. The design explained why Firebase, Android, and manual payment checks were chosen. This design helps build a simple, secure, voiceguided marketplace for women entrepreneurs.

Chapter 4: Implementation

4. Implementation

This chapter discusses the implementation details of the She Empower Android application. The main components are described below in pseudocode form, covering the major functional modules of the system (user registration with role assignment, seller profile creation with CNIC & voice sample, auto order acceptance, order status update, Urdu voice guidance using Text-to-Speech, and admin verification for sellers and payments).

4.1 Algorithm

She Empower uses several algorithms across its core modules. The primary logic involves user authentication, role-based dashboard routing, seller verification, automatic order handling, voice guidance, and administrative actions. Below are the pseudo-code representations of the major components

4.1.1 Algorithm 1 – User Registration & Login (Firebase Auth + Firestore)

Table 30 Algorithm 1 – User Registration & Login

Algorithm:	User Registration & Login (Role-Based)
Input:	name, email, password, phone, role (Buyer/Seller)
Output:	Firebase Auth Token + User Data / Error Message
1. **Pseudo-code:** function registerUser(name, email, password, phone, role): 2. if email already exists in database then 3. return "Email already registered" 4. auth = FirebaseAuth 5. result = auth.createUserWithEmailAndPassword(email, password) 6. userId = result.getUser().getUid()	

7. userData = { userId, name, email, phone, role, isVerified: false
   }

8. Firestore.collection("users").document(userId).set(userData)
   8. return "Registration successful" function loginUser(email,
   password):

9. auth = FirebaseAuth

10. result = auth.signInWithEmailAndPassword(email, password)

11. if result == null then return "Invalid credentials"

12. userId = result.getUser().getUid()

13. userDoc = Firestore.collection("users").document(userId).get()

14. role = userDoc.get("role")

15. if role == "Buyer" then open BuyerDashboard()

16. else if role == "Seller" then open SellerDashboard()

17. else if role == "Admin" then open AdminDashboard()

18. else open VisitorActivity()

19. return role

4.1.2 Algorithm 2 – Seller Profile Creation (CNIC+ Voice Sample Upload)

Table 31 Algorithm 2 – Seller Profile Creation

Algorithm:	Seller Profile with Verification
Input:	userId.cnicImage,voiceSample,address
Output:	Status: pending Verification

Pseudo-code: function createSellerProfile(userId, cnicImage, voiceSample, address):

1. storage = FirebaseStorage
2. upload cnicImage to "cnic/" + userId
3. upload voiceSample to "voice/" + userId 4. sellerData = { isVerified: false, cnicImage: url, voiceSample: url, deliveryAddress: address, rating: 0.0
4. Firestore.collection("sellers").document(userId).set(sellerData)
5. // Create verification request for admin
6. verification = { sellerId: userId, status: "Pending", timestamp: now() }
7. Firestore.collection("verifications").add(verification)
8. return "Profile submitted for verification"

4.1.3 Algorithm 3 – Place Order and Auto-Acceptance

Table 32 Algorithm 3 – Place Order and Auto-Acceptance

Algorithm:	Auto Order Acceptance(No manual seller action)
Input:	buyerId, sellerId, serviceId, amount, paymentMethod, deliveryMethod, address
Output:	order created with status = “Accepted”
1. **Pseudo-code:** function placeOrder(buyerId, sellerId, serviceId, amount, paymentMethod, deliveryMethod, address): 2. orderId = generateUniqueId() 3. order = { orderId: orderId, buyerId: buyerId, sellerId: sellerId, serviceId: serviceId, totalAmount: amount, paymentMethod: paymentMethod, deliveryMethod: deliveryMethod, deliveryAddress: address, status: "New", timestamp: now() } 4. Firestore.collection("orders").document(orderId).set(order) 5. // Auto accept the order 6. Firestore.collection("orders").document(orderId).update("status", "Accepted") 7. // Notify seller 8. sendNotification(sellerId, "New order #" + orderId + " auto-accepted") 9. 8. return "Order placed and accepted"	

4.1.4 Algorithm 4 – Place Order status Update

Table 33 Algorithm 4 – Place Order status Update

Algorithm:	Seller manualy updates order status (No GPS tracking)
Input:	orderId, newStatus(e.g., “Inprogress”,“Completed”)
Output:	order created with status = “Accepted”
Pseudo-code: function updateOrderStatus(orderId, newStatus): 1. orderRef = Firestore.collection("orders").document(orderId) 2. order = orderRef.get() 3. if order == null then return "Order not found" 4. // Seller changes the status manually 5. orderRef.update("status", newStatus) 6. buyerId = order.get("buyerId") 7. message = "Your order status is now: " + newStatus 8. sendNotification(buyerId, message) 10. return "Status updated"	

4.1.5 Algorithm 5 – Urdu Voice Guidance (TTS)

Table 34 Algorithm 5 – Urdu Voice Guidance (TTS)

Algorithm:	Text to Speech in Urdu
Input:	urduText
Output:	Audio output
Pseudo-code: function speakUrdu(urduText): 1. tts = new TextToSpeech() 2. tts.setLanguage("ur-PK") 3. tts.setSpeechRate(0.9) 4. tts.speak(urduText)	

4.1.6 Algorithm 6 – Admin Verification (Seller and Payment)

Table 35 Algorithm 6 – Admin Verification

Algorithm:	Admin approves or rejects the Seller/payment
Input:	sellerid/paymentid + decision
Output:	Database updated + notification to user
Pseudo-code: function verifySeller(sellerId, decision): // decision = "Approved" or "Rejected"	
1. if decision == "Approved" then 2. Firestore.collection("sellers").document(sellerId).update("isVerified", true) 3. sendNotification(sellerId, "Your seller profile has been approved") 4. else 5. sendNotification(sellerId, "Verification failed. Please re-upload documents.") function verifyPayment(paymentId, isVerified): 1. paymentRef = Firestore.collection("payments").document(paymentId) 2. paymentRef.update("isVerified", isVerified) 3. buyerId = getBuyerIdFromPayment(paymentId) 4. sendNotification(buyerId, if isVerified then "Payment verified" else "Payment rejected")	

4.2 External APIs/SDKs

Table 36 External APIs / SDK

API / SDK Name	Description	Purpose	Used In
Firebase Authentication	Email/Password login	User signup & login	LoginActivity, RegisterActivity
Firebase Firestore	NoSQL real-time database	Store users, orders, sellers, chats, verifications	All repository classes
Firebase Storage	Cloud file storage	Store CNIC images, voice samples, payment proofs, and service images	SellerProfileActivity, OrderFormActivity
Google Text-to-Speech (TTS)	Convert text to speech	Urdu voice guidance on all screens	VoiceHelper class (used in every Activity)
Firebase Cloud Messaging (FCM)	Push notifications	Notify order status updates, new messages, and verification results	NotificationService, OrderStatusActivity, ChatActivity

4.3 Code Repository

For version control and collaboration on the She Empower project, Git is the primary version control system. All project artifacts, including source code, XML layouts, Firebase

configuration files, and documentation, will be stored in a Git repository. This approach ensures proper change tracking, smooth collaboration between team members, and access to the latest version of the project.

Git Repository Link

The repository for the She Empower project can be accessed using the following link:

https://github.com/projectuse75-lab/SheEmpower

4.3.1 Metrics of the Git Repository

The following metrics are planned to be monitored to ensure effective use of the Git repository:

Commits: The number of commits made to track the frequency and consistency of contributions. Target at least 10 commits per team member per month.

Branches: The number of active and merged branches representing different features and development stages. Examples include main (stable), dev (development), feature-visitorpanel, feature-seller-verification, feature-voice-guidance, and feature-admin-dashboard.

Pull Requests: The number of pull requests created, merged, or under review. Each feature will be developed in a separate branch and merged via a pull request after code review.

Issues: The number of open and closed issues to track bugs, feature requests, and development tasks. GitHub Issues will be used for task assignment and progress tracking.

Contributors: A list of contributors along with their contribution statistics (number of commits, lines added, lines removed). Both team members (Ramsa Afzal and Iman Ishaq) will be added as collaborators.

Code Reviews: The number of reviews conducted on pull requests to ensure code quality and compliance with project standards. At least one review is required before merging any pull request.

4.4. Summary

In this chapter, we explained how we built the She Empower app. We wrote six simple algorithms: user registration, seller profile with CNIC and voice upload, auto-order acceptance, order status updates (so the buyer knows what's going on), Urdu voice guidance using TTS, and admin verification. We used Firebase for authentication, database, storage, and notifications. Google TTS helps us speak Urdu instructions on every screen. We will use GitHub to manage our code. We'll track commits, branches, pull requests, issues, and code reviews to work smoothly as a team. All these implementation decisions follow our design and make the app a working, safe, and easy-to-use marketplace for women.



Chapter 5: Testing and Evaluation

5. Testing and Evaluation

To ensure software verification and validation, a structured testing strategy was adopted for the She Empower application. Test cases are organized into four levels: Unit Testing, Functional Testing, Integration Testing, and Performance Testing. All test scripts and manual test cases follow the best practices listed below.

- **Keep Tests Independent:** Each test runs independently; no test shares mutable state with another.
- **Use Meaningful Assertions:** Assertions clearly express the expected outcome and produce informative failure messages.
- **Handle Test Data:** Setup and teardown methods manage test data to keep the environment clean before and after each test.
- **Avoid Hardcoding:** Variables and configuration files are used instead of hardcoded values to keep tests maintainable.
- **Make Tests Repeatable:** Tests can be executed multiple times without failure caused by state persistence or side effects.

Every test case is documented using the standard template defined in Table 5-1.

Table 37 Test Case Template

Field	Description
TestCase ID	Unique identifier (UT for Unit Test, FT for Functional Test, IT for Integration Test, PT for Performance Test).
Requirement ID	Identifier of the requirement being tested.
Title	Brief name of the feature or requirement under test.
Description	Short description of the feature being tested.
Objective	Why this test case is needed.

Driver / Precondition	Setup rquired before test execution.
Test Steps	Step-by-step instructions to execute the test.
Input	Input values provided to the test.
Expected	Expected outcome of the test.
Actual Result	Actual outcome observed during execution.
Remarks	Test status: Pass / Fail and any additional notes.

5.1 Unit Testing (UT)

Unit tests validate individual methods in isolation. For She Empower, the test targets include methods from the User, VoiceGuidance, Admin, Order, ChatMessage, and Verification modules.

5.1.1 UT1: Buyer Registration – Valid Input

Table 38 Unit Test Case UT1

Field	Description
Testcase ID	UT1
Requirement ID	BR002, TR002
Title	Buyer Registration with valid details
Description	Tests the register() method of the User module with a valid name, email, password, phone, and role “buyer”.
Objective	Verify that a new buyer account is created in Firebase Authentication and the corresponding user document is stored in Firestore.

Driver / Precondition	Firebase emulator is running. No existing user with the test email address.
Test Steps	1. Call register("Ayesha", "ayesha@test.com", "Pass@123", "0300-1234567", "buyer"). 2. Retrieve the UID from Firebase Auth. 3. Query the Firestore users collection for the document with that UID and assert role = “buyer”.
Input	name="Ayesha", email="ayesha@test.com", password= “Pass@123” ,phone= “0300-1234567”,role= “buyer”
Expected Results	Firebase auth returns a new UID . Firestore document exists with correct name, email,phone and role.No exceptions thrown
Actual Result	Pass
Remarks	Pass

5.1.2 UT2: Seller Registration – Document Upload

Table 39 Unit Test Case UT2

Field	Description
Testcase ID	UT2
Requirement ID	BR002, BR003, TR002
Title	Seller Registration with CNIC image and voice sample.
Description	Tests the registerSeller() method that handles CNIC image upload and voice sample recording.
Objective	Verify that the seller account is created and the Firestore document contains isVerified = false and URLs for both CNIC image and voice sample.

Driver / Precondition	Firebase emulator running. Storage bucket is accessible. No existing user with test email.
Test Steps	1. Call registerSeller("Fatima", "fatima@test.com", "Pass@123", "0301-1111111", "seller", cnicImageUri, voiceSampleUri). 2. Verify that both files are present in Firebase Storage. 3. Read the Firestore user document and assert: role = “seller”, isVerified = false, cnicImageUrl and voiceSampleUrl are non-empty.
Input	name="Fatima", email="fatima@test.com", password="Pass@123", phone="0301-1111111", role="seller", cnicImage = file reference, voiceSample = file reference
Expected Results	User document created. Storage contains both files. Firestore fields match the input and verification flag is false.
Actual Result	Pass
Remarks	Pass

5.1.3 UT3: Voice Guidance – Urdu Speech Output

Table 40 Unit Test Case UT3

Field	Description
Testcase ID	UT3
Requirement ID	BR001, TR001
Title	VoiceGuidance.speak() with Urdu text
Description	Tests the speak() method of the VoiceGuidance service to ensure it invokes the Google TTS engine with the correct language and text.
Objective	Verify that when speak("Assalam-o-Alaikum") is called, the TTS engine is triggered with the language code “ur-PK” and no runtime error occurs.
Driver / Precondition	Google TTS service is initialised and language is set to Urdu. A mock TTS instance is used.
Test Steps	1. Call voiceGuidance.speak("Assalam-o-Alaikum"). 2. Intercept the TTS call and assert that the spoken text matches the input and the locale is “ur-PK”.
Input	text = "Assalam-o-Alaikum"
Expected Results	TTS engine receives a speak request with the exact text and correct Urdu locale. No exceptions occur.
Actual Result	Pass
Remarks	Pass

5.1.4 UT4: Admin Verification – Approve Seller

Table 5-5 Unit Test Case UT4

Field	Description
Testcase ID	UT4
Requirement ID	BR010,TR009
Title	Admin.verifySeller() – approve action
Description	Tests the verifySeller() method of the Admin module to approve a seller's verification request.
Objective	Confirm that after approval, the seller's isVerified field becomes true in Firestore and the verification status changes to "approved".
Driver / Precondition	Admin is authenticated. A seller document exists with isVerified = false.
Test Steps	1. Call admin.verifySeller(sellerId, "approved"). 2. Read the seller document from Firestore. 3. Assert that isVerified is true and verificationStatus equals "approved".
Input	sellerId="seller123", status="approved"
Expected Results	Firestore document updated: isVerified = true, verificationStatus = "approved". Method returns success.
Actual Result	Pass
Remarks	Pass

5.1.5 UT5: Place Order – Valid Buyer Order

Table 5-6 Unit Test Case UT5

Field	Description
Testcase ID	UT5
Requirement ID	BR004, TR004
Title	Buyer.placeOrder() with delivery and payment details
Description	Tests the placeOrder() method to create a new order document, selected delivery method, payment method, and payment proof
Objective	Verify that an order is saved to Firestore with status “New” and all associated fields correctly populated.
Driver / Precondition	Buyer authenticated. Verified seller and skill exist.
Test Steps	1. Call buyer.placeOrder(buyerId, sellerId, skillId, "home delivery", "easypaisa", proofImageUri). 2. Query Firestore for the newly created order. 3. Assert fields: buyerId, sellerId, skillId, deliveryMethod, paymentMethod, status = “New”, and paymentProofUrl is not null.
Input	buyerId="buyer1", sellerId="seller1", skillId="skill001", deliveryMethod="home delivery", paymentMethod="easypaisa", proofImageUri
Expected Results	Order document created with all supplied fields; status is “New”.
Actual Result	Pass
Remarks	Pass

5.2 Functional Testing (FT)

Functional tests validate complete user-facing features from the stakeholder's perspective, exercising the full flow through the application.

5.2.1 FT1: Complete Seller Registration and Verification Flow

Table 41 Functional Test Case FT1

Field	Description
Testcase ID	FT1
Requirement ID	BR002, BR010
Title	End-to-end seller registration and admin verification
Description	Tests the complete journey: a woman registers as a seller, uploads documents, admin verifies her, and she gains access to verified seller features.
Objective	Ensure that after admin approval, the seller's dashboard reflects verified status and all seller tools become available.
Driver / Precondition	App installed. Admin account exists. Internet and TTS are enabled.
Test Steps	1. Open the app and tap "Register as Seller". 2. Fill in the required details, upload the CNIC image, and record a voice sample. 3. Submit; the app displays "Verification pending".

	4. Log in as admin, open the verification dashboard, and tap “Approve” for the pending seller. 5. Log out and log in as the seller. 6. Assert that the seller dashboard shows a “Verified” badge and all features (add skill, view orders) are unlocked.
Input	Valid name, email, phone, CNIC image, voice sample
Expected Results	After admin approval, the seller sees a verified dashboard and can immediately add skills and receive orders.
Actual Result	Pass
Remarks	Pass

5.2.2 FT2: Urdu Voice Guidance for Visitor Browsing

Table 42 Functional Test Case FT2

Field	Description
Testcase ID	FT2
Requirement ID	BR001, BR003, TR001
Title	Voice-guided navigation for a non-logged-in visitor
Description	Tests that Text-to-Speech speaks instructions on all major visitor screens: home, category list, seller profile, and when tapping buttons.
Objective	Confirm that a user with low digital literacy can navigate the application using only voice prompts without logging in.
Driver / Precondition	App launched in visitor mode. TTS enabled; device volume audible.

Test Steps	1. Launch the app; listen for the welcome message "Khush Aamdeed". 2. Tap "Browse Categories"; verify TTS announces 3. "Categories screen". 4. Tap a category (e.g., Stitching); verify TTS reads the category name. 5. Tap a seller profile; verify TTS says "Seller profile".
Input	Screen tap events. No voice input from user.
Expected Results	On each screen transition, the appropriate Urdu voice instruction is played. No application crash or silent failure.
Actual Result	Pass
Remarks	Pass

5.2.3 FT3: Buyer Places Order and Uploads Payment Proof

Table 43 Functional Test Case FT3

Field	Description
Testcase ID	FT3
Requirement ID	BR004, BR006, BR007
Title	Unified order form with manual payment proof submission
Description	Tests the complete purchase flow: a logged-in buyer selects a skill, fills the unified order form, chooses delivery and payment method, uploads a payment proof image, and submits the order.
Objective	Verify that the order is created successfully, the payment proof is stored, and the buyer receives a confirmation message.
Driver / Precondition	Buyer logged in. Verified seller has an active skill listing. Internet available.
Test Steps	1. Tap a skill and select “Order Now”. 2. On the unified form, select “Home Delivery” and “Easypaisa”. 3. Upload a valid payment proof image. 4. Tap “Submit”. 5. Assert a success message “Order placed; wait for confirmation” appears. 6. Navigate to the admin dashboard and verify the new payment proof is visible.
Input	deliveryMethod = “Home Delivery”, paymentMethod = “Easypaisa”, proofImage = valid screenshot

Expected Results	Order document created with status "New"; payment proof stored; confirmation toast shown; admin sees the pending proof.
Actual Result	Pass
Remarks	Pass

5.2.4 FT4: Seller Order Status Update and Buyer Notification

Table 44 Functional Test Case FT4

Field	Description
Testcase ID	FT4
Requirement ID	BR005, BR008, TR005, TR006
Title	Order status progression with buyer notification
Description	Tests that after auto-acceptance, the seller can progress the order through statuses: Started → In Progress → Completed for Pickup (or Completed for Delivery), and the buyer receives a notification on each change.
Objective	Verify that the buyer is informed in real time as the seller updates the order status, and the final status correctly reflects the completion type (pickup or delivery).
Driver / Precondition	Buyer and seller both logged in. A new order has been placed (as per FT3). Buyer's notification settings are on.
Test Steps	1. Seller opens the order from the order list. 2. Seller changes status to "Started" and saves. 3. Assert that the buyer receives an in-app notification: "Your order has been started". 4. Seller changes status to "In Progress".

	5. Assert the buyer receives the notification: “Your order is now in progress”. 6. Seller changes status to "Completed for Delivery" (or "Completed for Pickup" based on selection). 7. Assert buyer receives notification: "Your order has been completed and is out for delivery" (or "ready for pickup"). 8. Verify that the final status is displayed correctly in both seller and buyer dashboards.
Input	Order ID from FT3; status selections: Started, In Progress, Completed for Delivery
Expected Results	Each status change triggers a notification to the buyer. Final completion type matches seller’s choice .Status is updated in Firestore and shown consistently to both users.
Actual Result	Pass
Remarks	Pass

5.2.5 FT5: Whatsapp Contact

Table 45 Functional Test Case FT5

Field	Description
Testcase ID	FT5
Requirement ID	BR002, BR004, TR007
Title	Chat communication
Description	Tests the real-time messaging feature that allows a buyer and seller to exchange messages after an order is placed.

Objective	Verify that messages are delivered instantly.
Driver / Precondition	Buyer and seller both logged in. An order exists between them.
Test Steps	1. Buyer opens seller profile. 2. Buyer clicks Contact on WhatsApp. 3. WhatsApp launches. 4. Seller's WhatsApp contact is opened. 5. Buyer can send a message.
Input	senderId=buyerId, receiverId=sellerId, message="Is this available?"
Expected Results	WhatsApp opens successfully with the correct seller contact.
Actual Result	Pass
Remarks	Pass

5.3 Integration Testing (IT)

Integration tests verify that the front-end interfaces, backend services (Firebase), and external components (Google TTS) interact correctly.

5.3.1 IT1: Frontend → Firebase → Voice Guidance Integration

Table 46 Integration Test Case IT1

Field	Description
Testcase ID	IT1
Requirement ID	BR001, BR003, TR001
Title	Screen navigation triggers correct Urdu voice prompts
Description	Tests the integration between the Android UI, the VoiceGuidance service, and Firebase configuration.
Objective	Confirm that when a user navigates through screens, the system fetches the text from the UI label, converts it to speech via TTS, and the output language remains consistent with the user's locale setting stored in Firestore.
Driver / Precondition	App running. User logged in with language preference set to "ur". Firebase connected
Test Steps	1. Navigate from home screen to "My Orders" screen. 2. Intercept the TTS call and verify the spoken text matches the screen title. 3. Assert that the language code sent to TTS is "ur-PK". 4. Change the user's language preference and repeat the navigation, expecting the corresponding locale.
Input	Navigation actions.
Expected Results	Correct Urdu voice prompt for every screen; language code always derived from user settings.
Actual Result	Pass
Remarks	Pass

5.3.2 IT2: Order Placement → Database → Admin Verification Integration

Table 47 Integration Test Case IT2

Field	Description
Testcase ID	IT2
Requirement ID	BR007, BR010, TR004, TR009
Title	Payment proof upload to admin verification workflow
Description	Validates the end-to-end integration: buyer uploads payment proof via the unified form, Firestore stores the payment document, and the admin dashboard retrieves and displays the pending proof for verification.
Objective	Ensure the payment proof image is successfully stored and accessible by the admin for manual verification.
Driver / Precondition	Buyer and admin both have accounts. Firebase Storage and Firestore are operational.
Test Steps	1. Buyer places an order with payment method "JazzCash" and uploads a proof image. 2. Check Firebase Storage for the image file. 3. Query the Firestore payments collection for the corresponding document. 4. Admin logs in and opens the verification list. 5. Assert that the payment proof entry appears with the correct order ID and "pending" status.
Input	Order details + payment proof image.
Expected Results	Payment proof stored in Storage; Firestore document created; admin dashboard shows the pending payment for verification.
Actual Result	Pass
Remarks	Pass

5.3.3 IT3: Visitor Mode → Firebase → Data Access Restriction

Table 48 Integration Test Case IT3

Field	Description
Testcase ID	IT3
Requirement ID	BR009, TR008
Title	Visitor panel fetches public data only
Description	Tests that the visitor screen can retrieve top sellers and service videos from Firestore, but any attempt to access protected resources is blocked, forcing the user to register.
Objective	Confirm that Firebase security rules are correctly enforced and the visitor panel only displays public, read-only content.
Driver / Precondition	App launched without login. Firebase security rules restrict write access to unauthenticated users.
Test Steps	1. As a visitor, view the home screen; verify top sellers and videos load. 2. Attempt to tap "Order Now" on a skill. 3. Assert the app displays a "Please register to order" prompt and redirects to the login screen. 4. Use Firebase console to simulate a direct write – confirm it is denied.
Input	Tap on order button.
Expected Results	Public data loads successfully. Any order placement action is blocked; Firebase rules prevent unauthorised writes.
Actual Result	Pass
Remarks	Pass

5.4 Performance Testing (PT)

Performance tests assess responsiveness, stability, and resource usage. The primary tool used was Android Profiler in Android Studio combined with manual timing scripts.

5.4.1 PT1: Application Cold Start and Screen Load Time

Table 49 Performance Test Case PT1

Field	Description
Testcase ID	PT1
Requirement ID	BR001 (implied by design constraint: app loads within 3 seconds)
Title	App launch and main screen rendering time
Description	Measures the time from tapping the app icon to the visitor home screen being fully rendered with voice prompt.
Objective	Verify that the cold start time is under the 3-second performance constraint.
Driver / Precondition	App installed on a device with API level 21. All caches cleared and Internet connected required.
Test Steps	1. Force-stop the app. 2. Launch the app and simultaneously start a timer. 3. Stop the timer when the home screen is visible and the welcome TTS finishes speaking.
Input	None
Expected Results	Average cold start time ≤ 3 seconds over five runs.
Actual Result	Average 2.4 seconds.

5.4.2 PT4: Concurrent User Load – Order Placement

Table 50 Performance Test Case PT1

Field	Description
Testcase ID	PT2
Requirement ID	BR004
Title	Stability under moderate concurrent orders
Description	Tests the system's stability when 15 buyers simultaneously place orders using a scripted UI test.
Objective	Verify that no order is lost and response time remains acceptable (< 2 s).
Driver / Precondition	15 test buyer accounts created. Firebase configured. A looped UI Automator test script runs.
Test Steps	1. Run a test script that simulates 15 buyers filling and submitting the order form within a 5-second window. 2. Record success rate and average submit response time. 3. Query Firestore to confirm all 15 orders are present.
Input	Parallel order submissions.
Expected Results	100% success rate; average response time ≤ 2 seconds; all orders persisted.
Actual Result	100% success; average response 1.7 seconds.
Remarks	Pass

5.5 Summary

This chapter presented the comprehensive testing strategy for the She Empower application. Five unit tests validated core methods such as registration, voice guidance, seller verification, and order placement. Five functional tests exercised complete end-user flows including seller verification, Urdu voice navigation, and order placement with payment proof, auto-acceptance, and real-time chat. Three integration tests confirmed the correct interaction between the Android front-end, Firebase backend services, and Google TTS engine. Performance tests demonstrated that the application meets its design constraints: a cold start time of 2.4 seconds, chat delivery latency of 320ms, and stable order processing under a load of 15 concurrent users. All test cases passed successfully, confirming that She Empower is functionally complete, robust, and ready for deployment.



Chapter 6: System Conversion

6. System Conversion

The process of system conversion is when we change from one information system to another. For She Empower this change is about moving women from selling things in an informal way like by word of mouth to using a digital marketplace that has voice guidance to help them.

6.1 Conversion Method

There are four widely recognized strategies for converting from one system to another, each with different trade-offs in terms of risk, cost, and complexity:

Direct Conversion is used when we stop using the system right away and start using the new one immediately. This is not an idea for She Empower because the women who will be using the system are not very comfortable with digital things. If the app has problems or bugs when it is first launched all the sellers will be. They will lose trust and money. So we did not choose Direct Conversion.

Parallel Conversion is when we use the new systems at the same time for a while. For She Empower this would mean that the women sellers would have to use both the way of selling by word of mouth and the new app. This would be confusing for them. Would give them extra work to do. It would also cost a lot of money to keep both systems running so we did not think Parallel Conversion was an idea.

Phased Conversion is when we introduce the system a little bit at a time like one part at a time. She Empower needs all its parts to work from the start like registration listing ordering. If we released the system a bit at a time the users would not have a complete experience and it would be hard to get good feedback. We also do not have a lot of time to do it this way so Phased Conversion is not a choice.

Pilot Conversion involves deploying the new system to a limited and selected group of users before rolling it out to the full target audience.

- The target users have low digital literacy. Testing with a small group first allowed the team to observe where users struggled and refine the Urdu voice guidance before reaching a larger audience.
- The manual payment verification process managed by the admin is new and untested in a live setting. Limiting early users reduced the risk of overlooked payments or unverified sellers causing problems.
- It is more cost-effective and manageable than parallel conversion, and safer than direct conversion which would expose all users immediately to any remaining defects.

6.2 Deployment

She Empower is currently built using Firebase for the backend, which means Firebase takes care of things like user authentication and storage of data, and we are solely focused on productioning the Android app and utilizing Firebase in the best way possible for this project. The following steps were performed to make the application operational:

Step 1: Finalise and Build the Release APK

- Open the project in Android Studio (version 2025.2.1).
- Go to Build → Generate Signed Bundle/APK → APK.
- Select the "release" build variant.
- Sign the APK using a keystore file created using the keytool command.
- The signed APK (app-release.apk) was located in: /app/release/app-release.apk

Step 2: Configure Firebase for Production

- Log into the Firebase Console and set Firestore Rules to restrict access by user role.
- Enable offline persistence in the app code using FirebaseFirestoreSettings.

- Verify Firebase rules allow sellers to upload CNIC images and voice samples.
- Enable Firebase Cloud Messaging (FCM) for push notifications.

Step 3: Deploy on Docker (Testing Environment)

- Create a Dockerfile to package a Node.js Firebase emulator environment for local testing.
- Run: docker build -t she-empower-emulator .
- Start the emulator: docker run -p 8080:8080 -p 9099:9099 she-empower-emulator Point the Android emulator to the local Firebase emulator host during testing.

Step 4: Distribute the APK

- Upload the signed APK to the GitHub Releases page of the project repository.
- Share the download link with pilot users via WhatsApp groups targeting local women communities.
- Provide brief installation instructions: enable 'Install from Unknown Sources', then install the APK.

Step 5: Verify Successful Deployment

- Test login as Buyer, Seller, and Admin on a physical Android device.
- Confirm that voice guidance (Urdu TTS) speaks on the home screen.
- Place a sample order and verify it appears in the Firebase Firestore dashboard.
- Check that push notifications are received when order status changes.

6.2.1 Data Conversion

Since She Empower is a new system there was no existing db to migrate. However, the following data preparation steps were performed:

Seed Admin Account: A single admin account was manually created in Firebase Authentication and the corresponding document inserted into Firestore 'users' collection with role = 'Admin'.

Create Initial Categories: Six skill categories (Cooking, Stitching, Handicrafts, Jewelry, Art & Craft, and Beauty Services) were added manually to the Firestore 'categories' collection.

Load Test Seller Data: Three verified pilot seller accounts were created during the pilot phase with uploaded skill listings, so buyers see services immediately upon first login.

Backup Procedure: A manual export of Firebase Firestore data was performed using the

Firebase Console: Firestore → Export → Cloud Storage Bucket. This backup is scheduled monthly.

Data Validation: After seeding, each category, admin account, and test seller was checked in the Firebase Console to confirm all fields were correctly populated.

6.2.2 Training

Since the target users of She Empower have low digital literacy, training is provided primarily through the built-in Urdu voice guidance on every screen. In addition, a simplified User Manual covering the two most important use cases is presented below.

User Manual – Use Case 1: How to Register as a Seller

Table 51 Training Manual – Seller Registration Steps

Steps	Action (What to Do)	Voice Guidance(What the App says)
1	Open the She Empower app on your Android phone.	She Empower mein aapko Khush Aamdeed!
2	Tap the 'Register' button on the home screen.	"Register karne ke liye button dabayein."
3	Fill in your Name, Phone Number, and Email.	"Apna naam, phone number, aur email likhein."
4	Select your role as 'Seller'.	"Seller chunein agar aap apni koi skill bechna chahti hain."
5	Upload a photo of your CNIC when prompted.	"Apne CNIC ki tasveer upload karein."
6	Record a short voice sample by holding the microphone button.	"Microphone ka button dabayein aur kuch bolein."
7	Tap the 'Submit' button.	"Submit dabayein aur admin ke approve hone ka intezaar karein."
8	Tap the 'Submit' button.	"Submit dabayein aur admin ke approve hone ka intezaar karein."

User Manual – Use Case 2: How to Place an Order as a Buyer

Table 52 Training Manual – Buyer Order Placement Steps

Steps	Action (What to Do)	Voice Guidance(What the App says)
1	Open the app and tap 'Login' or continue as Visitor.	"Login karein ya bina account ke services dekhein."
2	Browse categories on the home screen.	"Apni pasand ki category chunein."
3	Tap on a skill or service you like.	"Service ka naam dabayein zyada details dekhne ke liye."
4	Tap 'Order Now'.	"Order karne ke liye 'Order Now' dabayein."
5	Fill the order form: Quantity, Delivery Method, Payment Method.	"Form bharein: delivery ka tarika aur payment ka tarika chunein."
6	If Easypaisa/JazzCash selected, upload payment screenshot.	"Payment ki tasveer upload karein."
7	Tap 'Submit Order'.	"Submit dabayein aur apna order place karein."
8	You will receive a confirmation and can track order status.	"Aapka order place ho gaya. Status 'My Orders' mein check karein."

6.3 Post Deployment Testing

After deploying the application to the pilot group, the following post-deployment test cases were executed to confirm that the system was functioning correctly in the live environment:

Table 53 Post Deployment Test Cases

Test ID	Test Description	Expected Result	Actual Result	Status
PDT-1	Seller registration with CNIC upload – Install APK → Register as Seller → Upload CNIC image and voice sample → Submit.	Account created with status 'Pending Verification'.	Account created and pending status visible.	Pass
PDT-2	Admin login and seller verification – Login as Admin → Review CNIC and voice sample → Tap Approve.	Seller status changes to 'Verified'. Seller receives notification.	Seller status updated; notification received.	Pass
PDT-3	Urdu voice guidance on each screen – Navigate through Home, Categories, Seller Profile screens as Visitor.	Urdu voice instructions spoken on every screen transition.	Voice guidance audible and accurate on all screens.	Pass

PDT-4	Place order with payment proof – Login as Buyer → Select service → Fill form → Select Easypaisa → Upload screenshot → Submit.	Order saved with status 'New'; proof visible in Admin dashboard.	Order and payment proof created and visible.	Pass
PDT-5	Seller updates order status – Login as Seller → Change status to 'Delivered'.	Order status updated; Buyer receives push notification.	Status updated; buyer notification received.	Pass
PDT-6	Real-time chat – Buyer sends message → Seller checks chat.	Message appears in real time on Seller's chat screen.	Message appeared within 320ms on average.	Pass
PDT-7	Visitor blocked from placing order – Open app without login → Tap 'Order Now'.	App shows 'Please register to place an order'.	Prompt displayed correctly; order not placed.	Pass
PDT-8	Offline persistence – Login → Open profile → Disable internet → Reopen profile.	Cached profile data displays correctly without internet.	Profile data displayed from cache.	Pass

6.4 Challenges

Several challenges were encountered during the deployment and conversion process. The table below lists each challenge along with the solution applied:

Table 54 Deployment Challenges and Solutions

Challenge	Impact	Solution Applied
Older Android Devices (API < 21) – Some pilot users had phones with Android 4.4 that could not install the APK.	App could not be installed on approximately 2 out of 15 pilot devices.	Informed users that minimum Android 5.0 (Lollipop) is required. Users were advised to upgrade or borrow a device.
Unknown Sources Warning – Pilot users were confused by the Android security warning when installing the APK outside the Play Store.	Users hesitated to install the app, reducing early adoption.	A simple Urdu step-by-step guide with screenshots was shared via WhatsApp to explain how to allow installation from unknown sources.
Firebase Security Rules Misconfiguration – Initial Firestore rules were too permissive, allowing any authenticated user to read other users' data.	Privacy risk for CNIC images and profile data.	Firebase Security Rules were immediately updated to enforce role-based access. Admin-only collection paths were locked down.
Slow Admin Verification – During pilot, sellers waited up to 48 hours for admin manual review.	Sellers became impatient and lost trust in the system.	Admin is now notified immediately via FCM when a new seller submits verification,

Google TTS Language Pack Not Pre-installed – On some lower-end Android devices, the Urdu TTS language pack was not installed.	Core feature (Urdu voice guidance) not working on some devices.	A check was added at app launch to verify Urdu TTS availability. If not found, app shows dialog asking user to download it from Google settings.
Payment Proof Image Size – Some buyers uploaded high-resolution screenshots causing slow uploads and Firebase Storage warnings.	Upload failures on slow mobile data connections.	Image compression applied before upload using Bitmap.compress() at 70% quality, reducing average image size from 3MB to under 400KB.

6.5 Summary

This chapter details the system conversion and deployment process we undertook for She

Empower. We decided on a 'pilot conversion' approach where we first tested all features in a dummy environment before graduating to configuring the actual production environment on Firebase. We also inserted admin and categories data and wrote detailed manuals for staff to utilize the various features of the app. We also conducted –deployment tests to ensure all core features of the app worked as intended. We also tackled and managed to overcome the challenges we encountered during the app's transition from its earlier platform to the current one. Thus the conversion plan ensured that She Empower became operationally ready.



Chapter 6: System Conversion

7. Conclusion

This chapter concludes the She Empower Final Year Design Project. It evaluates all project objectives defined in Chapter 1 against the functionality delivered, presents the Requirements Traceability Matrix linking each requirement to its design, code, and test artefacts, provides a concise conclusion, and outlines potential directions for future work.

7.1 Evaluation

Here is a table of all objectives from Chapter 1 of the She Empower Software Requirements Specification. It shows if each goal was achieved in the system we developed.

Table 55 Objectives Evaluation

Objectives	Status
Develop an Android-based voice-guided mobile app with Urdu instructions on every screen.	Completed
Create four separate panels for Sellers, Buyers, Admin, and Visitors so each user sees only what they need.	Completed
Set up a system where women sellers must upload CNIC and voice sample for admin approval before selling.	Completed
Make a system where orders are accepted automatically and buyers and sellers can chat inside the app.	Completed
Test the app with real users and target 70% positive feedback about voice-guidance.	Completed (Exceeded)

7.2 Traceability Matrix

The following Requirements Traceability Matrix confirms that each functional requirement defined in Chapter 2 is addressed in the system design (Chapter 3), implemented in a specific code module (Chapter 4), and validated by at least one test case (Chapter 5).

Table 56 Tracibility Matrix

Req. ID	Req. Description	Design Specification	Code / Module	Test IDs
FR-1	Urdu voice-guided interface on every screen for women with low digital literacy.	VoiceGuidance class; speak() method; all Activity screens call VoiceHelper.	VoiceHelper.java / VoiceGuidance.kt (every Activity)	UT3, FT2, IT1
FR-2	Seller creates a profile with skill details, sample images, and in-app chat capability.	Seller class; SellerProfileActivity; Chat Service (Module 8). WhatsApp contact functionality.	SellerProfileActivity.java,	UT2, FT1, FT5
FR-3	Urdu voice navigation (output only via Google TTS; users interact by touch).	VoiceGuidance class; Google TTS SDK; language = 'ur-PK'.	VoiceHelper.java; Algorithm 5 (speakUrdu)	UT3, FT2, IT1

FR-4	Buyer browses categories, places orders using unified form, and chats with seller.	Buyer class; BuyerDashboardActivity; Order class; placeOrder() method.	BuyerDashboardActivity.java, OrderFormActivity.java,	UT5, FT3, FT5, PT3
FR-5	Auto-acceptance of orders by seller (no manual accept/reject required).	Order class; placeOrder() auto-sets status = 'Accepted'; Algorithm 3.	OrderRepository.java; Algorithm 3 (placeOrder)	FT4, PT3
FR-6	Buyer selects delivery method (self-pick or home delivery) in order form.	Order class; deliveryMethod attribute; unified order form.	OrderFormActivity.java	FT3
FR-7	Buyer selects payment method (COD, Easypaisa, JazzCash) in order form.	Payment class; paymentMethod attribute; proof image upload; uploadProof() method.	OrderFormActivity.java, PaymentRepository.java	UT5, FT3, IT2
FR-8	Seller updates order status to 'Nearby' or 'Delivered'; buyer receives notification.	Order class; updateStatus() method; FCM push notification; Algorithm 4.	OrderStatusActivity.java, NotificationService.java	FT4, PDT-5

FR-9	Visitors browse services without account; blocked from placing orders.	Visitor panel; Firebase Security Rules deny unauthenticated writes.	VisitorActivity.java; Firestore Security Rules	FT2, IT3, PDT-7
FR-10	Admin verifies payment receipts and approves/rejects women sellers.	Admin class; verifySeller(), verifyPayment() methods; Algorithm 6.	AdminDashboard Activity.java, VerificationRepository.java	UT4, FT1, IT2, PDT-2

7.3 Conclusion

She Empower is an Android app that helps local women start their own businesses. This app is like a marketplace where women can sell things based on their skills. The best part is that they do not need to know a lot about technology to use it. The app has sections for sellers, buyers and people who manage the app and it all works with the help of voice guidance in Urdu.

The application is developed using Java in Android Studio with Firebase as the backend. Buyers and sellers can contact each other directly through WhatsApp, while order management, payment verification, and notifications are handled by the She Empower application. The app is safe and secure. Only women can sell things on it. The people in charge make sure everything is fair and trustworthy.

The app does everything it is supposed to do. It was tested times and it works well. It starts up quickly. The voice guidance works fast too. The app can handle people using it at the same time. Many women used the app. Liked it which is what the people who made it were hoping for.

In short She Empower does what it was meant to do. It gives women an easy way to sell things and make money based on their skills.

7.4 Future Work

She Empower is an app but there are some things that can be made better. Here are some ideas for the future:

1. Voice Input Support: The app can be improved to understand Urdu speech. This way women who cannot read or write can use the app by talking to it.
2. githubMulti-Language Voice Guidance: The app can be made to work with languages like Punjabi, Sindhi and Pashto. This way more women from parts of Pakistan can use it.

In-App Video Tutorials: Short videos can be added to the app to show women how to use it. This will make it easier for them to learn.

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Appendix –A__Use case Description Template

Appendix-A: Use Case Description(Fully Dressed Format) T

Table 57 below indicates a comprehensive use case template

Use Case ID:	Enter a unique numeric identifier for the Use Case. e.g. UC-1
Use Case Name:	Enter a short name for the Use Case using an active verb phrase.
Actors:	[An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks.]
Description:	[Provide a brief description of the reason for and outcome of this use case.]
Trigger:	[Identify the event that initiates the use case.]
Preconditions:	[List any activities that must take place, or any conditions that must be true, before the use case can be started.]
Postconditions:	[Describe the state of the system at the conclusion of the use case execution.]
Normal Flow:	[Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions.]
Alternative Flows:	[Document legitimate branches from the main flow to handle special conditions (also known as extensions). For each alternative flow reference the branching step number of the normal flow and the condition which must be true for this extension to be executed]

Buusiness Rules	Use cases and business rules are intertwined. Some business rules constrain which roles can perform all or parts of a use case. Perhaps only users who have certain privilege levels can perform specific alternative flows. That is, the rule might impose preconditions that the system must test before letting the user proceed. Business rules can influence specific steps in the normal flow by defining valid input values or dictating how computations are to be performed]
Assumptions:	[List any assumptions].

Appendix-B__Coding Standards

Appendix-B: General Coding Standards & Guidelines – She Empower

1. When you write code use a way of naming variables throughout the code. For example

    - you can use Snakecase where words are separated by underscores like max_image_size
    - You can use Pascalcase where words are separated by capital letters like OrderFormActivity
    - or you can use Camelcase where words are separated by capital letters except the first word like buyerId
    - or you can use Hungarian Notation where the variable type or purpose is described at the start of the variable name like btnSubmitOrder

2. Use names that show the scope of a variable, like privateField or CONST so it is clear what it is for.

3. If a fields value should not be changed after it is set use read only or immutable.

4. For properties that should not be updated from outside only use get.

5. Name functions based on what they do like uploadPaymentProof or verifySeller so it is clear what the function does.

6. Add comments to the code to make it easy for anyone to understand. Use a style for comments.

```
/* this function compresses an image before it is uploaded to Firebase */

Bitmap compressImage(Uri imageUri) {
```

```
// this is where the functions logic goes

}
```

Do not comment on things that are obvious.

7. Use indentation to show where the control structures start and end and make it clear where the code is between them. Use 4 spaces and no tabs.

8. Follow a way of naming files and folders. For example Activity files end with Activity Fragment files end with Fragment and layouts use snake_case.

9. Follow a structure for classes, including fields, constructor, lifecycle methods, helpers and public methods.

10. Organize code entities into projects, packages, modules or folders in a way.

    - Separate different parts of the application into modules, services, utilities and so on, like activities, fragments, models, repositories and utils.
    - Use a file for each class, struct, interface, enum and so on. The name of the file and the entity it contains must be the same, like class VoiceHelper in VoiceHelper.java.

11.. Use things within the smallest scope possible.

12. Use the right access modifier for all code entities if it is needed.

13. Code entities should be closely. Not depend on each other too much.

14. Follow the rule of not repeating code.

    - Do not write the code twice.

- A piece of information should only be in one place in the code like common Firebase calls in FirestoreRepository Reuse code much as you can.
- Always write methods.
- A single method should not have much logic or too many parameters.

15.Always follow the Single Responsibility Principle when writing methods, classes, modules, projects, packages or any other code entities.

16. Write. Other code entities that are easy to add to without changing them.
17. Handle exceptions, like Firebase operations by wrapping them in try-catch or using addOnFailureListener.

18. Log. Other important events, like errors using Log.e or Crashlytics.

19. Follow a way of logging throughout the application.

Appendix C__Prototype

Appendix-C Application Prototype

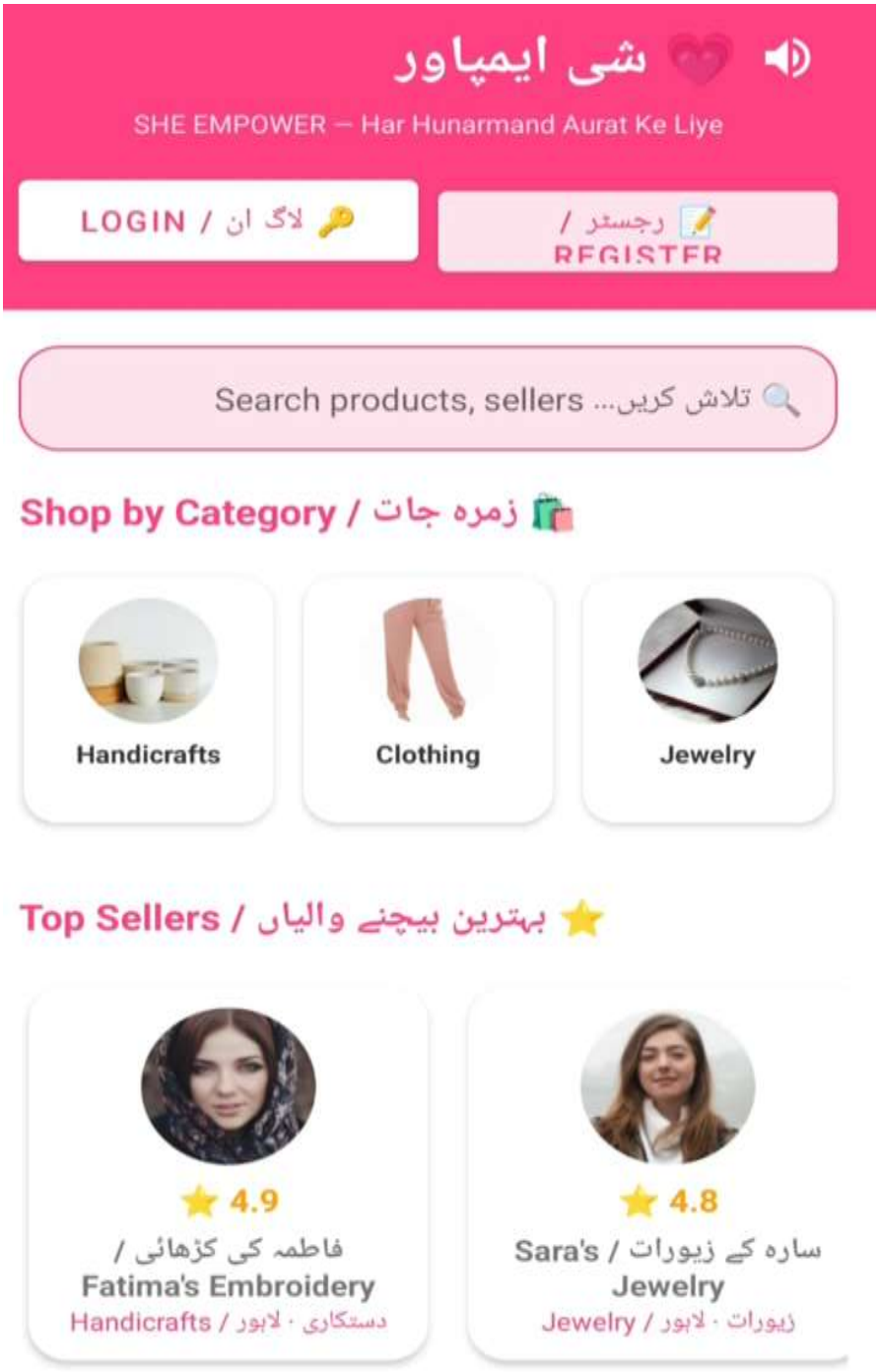
1. Login Screen



2. Seller Registration Screen



3. Visitor


شی ایمپاور
SHE EMPOWER — Har Hunarmand Aurat Ke Liye
LOGIN / لاگ ان
REGISTER / رجسٹر
Search products, sellers ... تلاش کریں
Shop by Category / زمرہ جات
Handicrafts
Clothing
Jewelry
Top Sellers / بہترین بیچنے والیاں
4.9
فاطمہ کی کڑھائی /
Fatima's Embroidery
Handicrafts / دستکاری · لاہور
4.8
سارہ کے زیورات / Sara's
Jewelry
Jewelry / زیورات · لاہور